



AI smart cane technology and assistive navigation for visually impaired users: an overview

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Abstract

Smart cane technologies have evolved into transformative tools enhancing mobility, safety, and autonomy for individuals with visual impairments. Despite rapid advances, comprehensive surveys are lacking that analyze the evolution of technology, the challenges of integrating these systems, and the design considerations. The objective of this study is to fill that gap by emphasizing the synergy of perception, decision-making, and feedback in smart cane systems, enabled through the integration of sensors, artificial intelligence, and human-centered design, including sensor-rich systems that incorporate Simultaneous Localization and Mapping (SLAM), computer vision, and machine learning techniques. Multimodal feedback is used to provide users with real-time localization, contextual awareness, dynamic obstacle avoidance, and personalized navigation based on their preferences. Several key hardware components are reviewed in the study, including ultrasonic, infrared, LiDAR, RGB-D cameras, and inertial sensors, and sensor fusion strategies are discussed to produce robust spatial perception results. A number of human-centered factors, such as usability, cognitive load, and ergonomics, must be considered in order to assure practical adoption. An integrated smart cane framework is presented that combines edge artificial intelligence, Bluetooth-based indoor localization, 5G/6G secure networking, cloud-based health monitoring, and solar-powered energy management. As a result of this architecture, scalable, intelligent mobility aids are capable of being seamlessly integrated into smart city infrastructures. The paper advocates for a multidisciplinary, systems-level approach to position smart canes as core enablers of inclusive, connected urban environments.

Keywords Smart cane · Assistive navigation · Sensor fusion · SLAM · AI-based navigation · Visually impaired mobility

1 Introduction

The global prevalence of vision loss is expected to rise dramatically in the coming decades, with projections from the International Agency for the Prevention of Blindness (IAPB Vision Atlas) indicating an increase from 1.1 billion affected individuals in 2020 to 1.8 billion by 2050—a 55% surge. This includes people experiencing blindness, as well as moderate to severe, mild, and near vision loss. Notably, women are anticipated to remain disproportionately affected, with 978 million cases compared to 780 million among men by 2050 (Anna Fleck 2024). This growing demographic underscores the urgent need for advanced assistive technologies that promote safe, autonomous mobility in increasingly complex urban environments. Among these innovations, smart canes integrated with intelligent navigation systems represent a major advancement over traditional white canes. Through the combination of sensors, Artificial Intelligence (AI), and

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real-time feedback mechanisms, these devices allow users to interact more effectively with their surroundings than with traditional white canes (Soares et al. 2021). There is a growing need for smart canes that provide dynamic obstacle detection, real-time route planning, and context-aware navigation in both indoor and outdoor environments as cities become increasingly complex. As such, technologies that address these challenges will be able to close critical accessibility gaps and allow visually impaired persons to move more confidently and safely (Beingolea et al. 2021). In this rapidly growing population, assistive technologies are critical to improving mobility and autonomy. There is an urgent need for advancements in assistive technologies that can enhance the quality of life and support independent mobility.

Traditional navigational aids such as the white cane have played a vital role for a long time; however, their utility is limited to detecting nearby obstacles based on physical contact. Despite their limited situational awareness, they are not capable of anticipating changes in the environment. Recently, smart canes equipped with Artificial Intelligence (AI) and Internet of Things (IoT) capabilities have been developed to overcome these limitations (Zafar et al. 2022). As an example, IoT-enabled systems are capable of collecting sensory data and relaying it to cloud-based platforms for real-time processing and adaptive guidance (Zafar et al. 2024). According to Gill et al. (J. Li et al. 2023), Multisensory assistive devices have the potential to monitor gait and enhance mobility based on data obtained from sensors. A smart navigation aid that recognizes objects and responds accordingly has been made possible by integrating components such as Raspberry Pi, cameras, ultrasonic sensors,

and speakers. The WeWALK Smart Cane offers enhanced navigational assistance by combining ultrasonic sensors with smartphone connectivity. Through these advancements, users are able to navigate modern environments with greater autonomy, safety, and quality of life (Suman et al. 2022).

This survey offers a unique and comprehensive synthesis that bridges these fragmented domains, merging advances in sensor fusion, AI-driven localization, human ergonomics, and system scalability. By systematically evaluating both technological and human factors, we aim to lay a foundation for the next generation of smart mobility aids that are robust, adaptive, and aligned with inclusive urban infrastructure. Our work stands apart by not only reviewing isolated components but also proposing a unified, future-oriented framework for AI-powered, cloud-connected, and ergonomically designed smart cane systems Fig. 1.

1.1 Motivations

Despite significant progress in developing smart cane systems, persistent limitations remain in terms of technology and from a user experience perspective (Nataraj et al. 2024). These challenges hinder the seamless adoption of smart canes and affect their real-world usability, particularly in diverse and dynamic environments. As shown in Fig. 2, current smart cane architectures integrate multiple sensors—including GPS, RGB-D cameras, IMUs, and LiDAR—to support functionalities such as obstacle detection, localization, and feedback delivery. Modern systems increasingly adopt Simultaneous Localization and Mapping (SLAM) algorithms to construct and interpret environmental maps in real time, which are essential for navigating unfamiliar or unstructured settings.

Fig. 1 Illustration of Smart Cane Architecture and SLAM Integration. (a) Laser SLAM: A LiDAR-based simultaneous localization and mapping system used for accurate obstacle detection and environment mapping in dynamic settings. (b) A system diagram of a smart cane shows sensor components, processing units, and output interfaces used to enable real-time navigation and user feedback

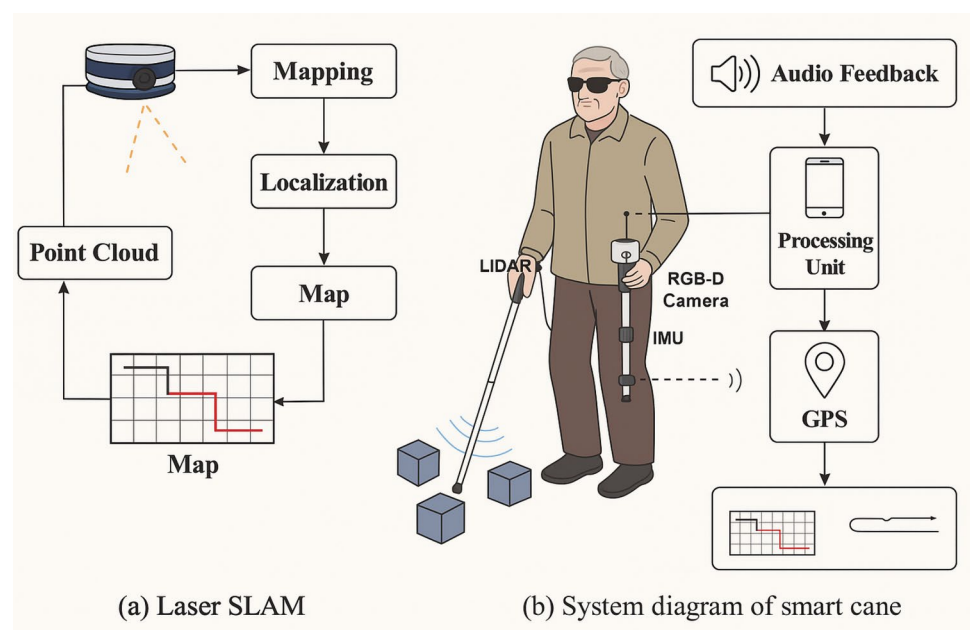


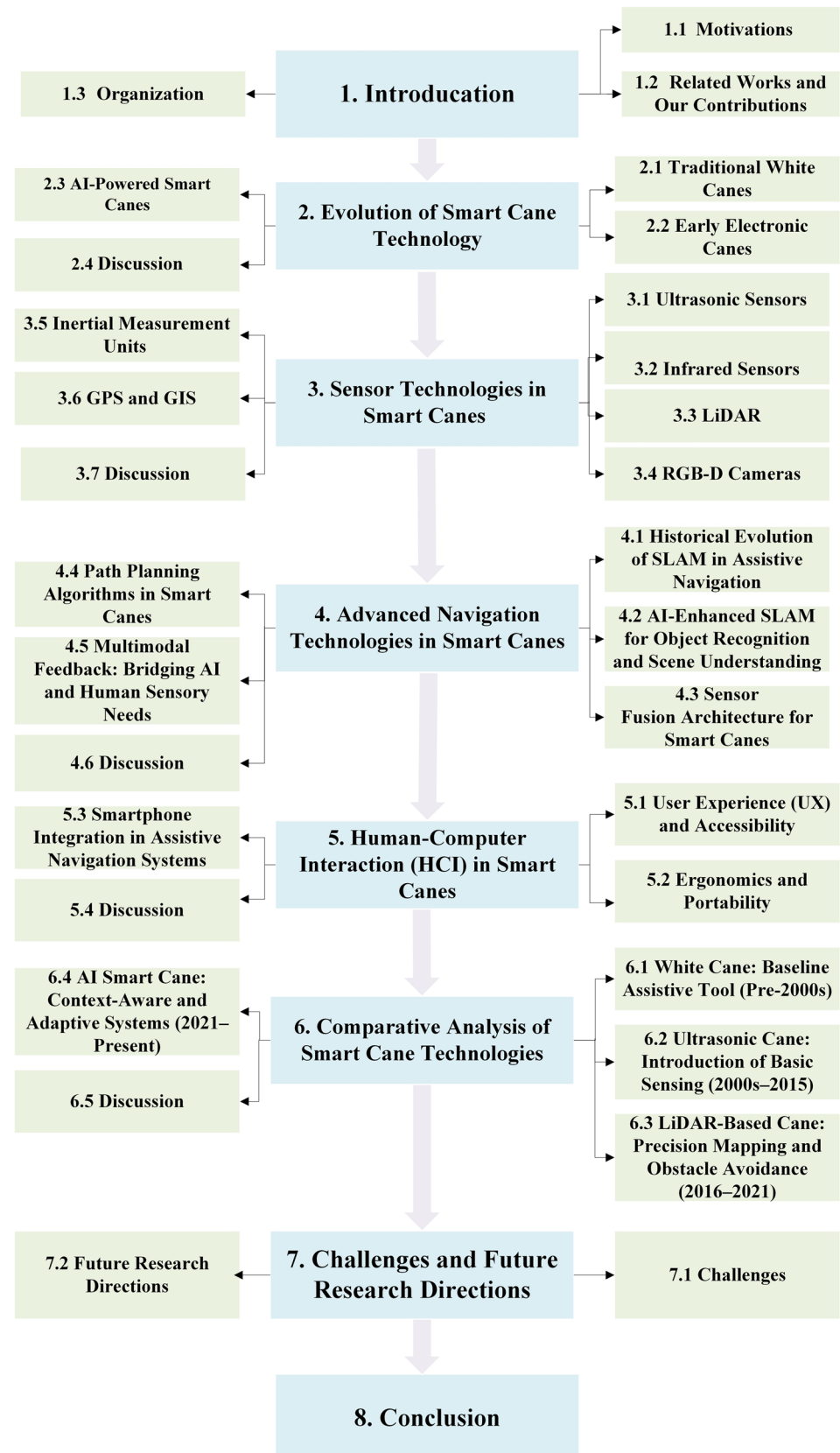
Fig. 2 The structure of the survey

Figure 2 presents an integrated overview of the technological and human-centric limitations associated with current smart cane systems. The diagram highlights three critical domains of concern: (a) navigation challenges across indoor and outdoor environments, (b) limitations in current cane technology, and (c) human factors affecting usability and acceptance. Each of these is elaborated below.

i. Outdoor vs. Indoor Navigation Limitations

- *GPS Inefficiency Indoors:* The Global Positioning System (GPS) is widely used for outdoor navigation (Ahmed et al. 2022). However, it is inefficient indoors due to signal attenuation and interference from multiple path signals. Due to this limitation, visually impaired individuals are hindered from navigating complex indoor spaces on their own (Gad et al. 2023).
- *Lack of Real-Time Localization Accuracy:* Providing navigation assistance in real time requires accurate real-time localization (Sudaraka Keshara et al. 2022). Achieving high precision in dynamic environments, such as areas with weak or unavailable GPS signals, remains a challenging task (Soares et al. 2021).

ii. Current Cane Limitations

- *Basic White Canes Offer Only Tactile Feedback:* A traditional white cane provides limited information because it relies entirely on tactile feedback for detecting obstacles within its reach. The impact of this constraint necessitates the development of more advanced solutions in order to improve the awareness of the environment (Gad et al. 2023; Soares et al. 2021).
- *Electronic Canes Rely on Single-Sensor Technology:* The majority of electronic canes are equipped with a single type of sensor, such as an ultrasonic or infrared one, which may not provide comprehensive environmental information. Through the incorporation of multiple sensors using sensor fusion techniques, obstacle detection (e.g., RGB-D cameras) and navigation accuracy can be enhanced (Mai et al. 2023).

iii. Human Factors in Assistive Navigation

- *Cognitive Load in AI-Powered Navigation:* Despite the advanced features that AI-enhanced navigation systems offer, they may increase cognitive load for users, potentially resulting in confusion or decreased usability. The mitigation of this issue requires the development of intuitive user interfaces and the provision of clear, concise feedback (Okolo et al. 2024).

- *User Acceptance & Ergonomics of Smart Canes:* The adoption of smart canes is dependent on a variety of factors, including ergonomics, weight, and ease of use. Providing these devices with comfort and ease of use is critical to their widespread acceptance among visually impaired individuals (Shahzad Chaudary 2022).

In parallel, human factors play a critical role in the adoption and usability of assistive navigation systems. Cognitive load, interface complexity, and ergonomic considerations all influence user acceptance. Smart cane technologies must, therefore, evolve in functionality and design, prioritizing user comfort, adaptability, and intuitive interaction. The number of research and development efforts in smart cane technologies and assistive navigation systems has increased significantly (Khan et al. 2021a, b; Mai et al. 2023; Messaoudi et al. 2022; Zafar et al. 2022). A comprehensive survey that synthesizes these advancements into a cohesive analysis remains lacking. The majority of review articles on assistive technology focus on particular aspects, such as sensor applications or AI methodologies, without considering the full scope of innovations in smart cane development.

1.2 Related works and our contributions

The existing surveys offer valuable insight into specific assistive technology areas such as SLAM (simultaneous localization and mapping), vision-based navigation, and sensor technology. However, as shown in Table 1, none of these technologies offer a comprehensive synthesis that combines all essential components, such as sensor fusion, artificial intelligence, and real-time localization, in one coherent system. The implementation of navigation technologies for visually impaired persons has been the subject of a number of existing surveys, but there remains a significant gap in providing a comprehensive and integrative review of these technologies. For instance, some studies focus narrowly on specific technologies such as SLAM-based navigation (Bamdad et al. 2024; Tezerjani et al. 2024) or AI and vision-based methods (Li et al. 2023), providing valuable insights but limiting their application. In contrast, others focus on a particular sensor technology, such as laser and vision sensors in smart canes (Mai et al. 2023), without exploring their integration with AI or user experience. The scope of some studies has been limited to SLAM-based navigation (Zafar et al. 2022), offering a comparative analysis of state-of-the-art tools, which often miss the connection between sensing technologies and real-world navigation performance. The review of navigation assistive tools notably lacks in-depth discussion on sensor fusion and AI-driven localization, which are critical for next-generation smart cane systems (Messaoudi et al.

Table 1 Comparison of related works

Title	Focus	Year
SLAM for Visually Impaired People: A Survey	Systematic review of SLAM-based solutions for blind and visually impaired navigation	2022
A Survey on Reinforcement Learning Applications in SLAM	An overview of Reinforcement Learning's utilization in SLAM reveals significant advancements in the field	2024
A Survey on Recent Advances in AI and Vision-Based Methods for Helping and Guiding Visually Impaired People	Overview of AI and vision-based assistive technologies for visually impaired individuals	2022
Laser Sensing and Vision Sensing Smart Blind Cane: A Review	Review of laser and vision sensing technologies in smart canes for the blind	2023
Assistive Devices Analysis for Visually Impaired Persons: A Review on Taxonomy	presents a comprehensive and comparative analysis of the state-of-the-art assistive devices for VIPs	2022
Review of Navigation Assistive Tools and Technologies for the Visually Impaired	General review of assistive navigation tools; lacks in-depth coverage of sensor fusion and AI-based navigation	2022
Analysis of Navigation Assistants for Blind and Visually Impaired People: A Systematic Review	Systematic review of navigation assistants, with a focus on usability and technological features	2021

2022). Furthermore, the systematic review on navigation assistants (Khan et al. 2021a, b), emphasizes usability and features yet does not comprehensively address technical limitations or indoor-outdoor adaptability. These fragmented perspectives highlight the need for a comprehensive survey that synthesizes recent technological advancements, such as sensor fusion, AI-based navigation, and human-centric design, and addresses current limitations and user needs in real-world scenarios. The motivation for this work is thus firmly rooted in bridging these gaps to inform future development and deployment of more effective assistive technologies.

Over the past decade, smart cane technology has evolved into a transformative solution that significantly enhances the mobility, safety, and autonomy of visually impaired individuals. While a growing body of literature has addressed various components of assistive navigation, such as sensor-based localization, computer vision, and AI-driven path planning, most studies remain fragmented, lacking a unified and forward-looking perspective. This comprehensive survey addresses that gap by offering the following key contributions:

- We present an overview of the technological progression in smart canes, tracing the transition from basic obstacle detection tools to advanced systems integrating real-time contextual awareness, adaptive navigation, and intelligent interaction. This historical mapping provides a foundation for understanding the pace and direction of innovation in assistive mobility.
- A detailed evaluation of multi-sensory integration strategies of the smart cane is conducted, focusing on how diverse sensory inputs are fused to enhance spatial awareness, obstacle detection, and system robustness.

Comparative insights into each modality's strengths and limitations in dynamic and unstructured environments are provided.

- We analyze state-of-the-art AI methodologies employed in smart cane systems, including Simultaneous Localization and Mapping (SLAM), object detection models like YOLO, and learning architectures such as CNNs and R-CNNs. These methods are reviewed for their ability to support real-time localization, semantic understanding, and user-centered path planning.
- Recognizing the importance of user experience, we examine ergonomic design, cognitive load, interface accessibility (e.g., multilingual voice and haptic feedback), and social acceptability. These factors are crucial for long-term adoption and personalization of assistive devices.
- The paper identifies persisting challenges, such as power efficiency, cost scalability, interoperability, and navigation in cluttered or GPS-denied environments. Based on these insights, we recommend research trajectories that include secure edge–cloud collaboration, three-dimensional environmental mapping through enhanced sensor fusion, and the development of universal design and accessibility standards for global deployment.
- Building upon current trends, we propose an integrated, scalable framework that envisions smart canes as edge-AI-powered, cloud-connected systems within smart city ecosystems. Key innovations include 5G/6G-enabled networking, secure BLE-based indoor localization, adaptive Machine Learning (ML) modules, and renewable energy integration (e.g., solar). This future-oriented model emphasizes modularity, security (via data encryption), and sustainable design, aligning with inclusive urban mobility goals.

This survey is both timely and valuable, given the advancements in assistive technologies and the need for comprehensive reviews in this domain. While some surveys have focused on specific aspects, such as SLAM for visually impaired navigation (Bamdad et al. 2024; Tezerjani et al. 2024), there is a lack of comprehensive surveys encompassing the broader scope of smart cane technologies, including sensor fusion and AI-based navigation. The existing surveys address specific components of assistive navigation technologies (Jain and Patel 2023; Khan et al. 2021a, b; Messaoudi et al. 2022; Zafar et al. 2022) but do not provide a holistic view of smart cane technologies integrating sensor fusion, SLAM, and AI-based navigation. As a result of this survey, we can bridge this gap by providing a comprehensive examination of these integrated systems, evaluating their effectiveness, usability, and impact on the mobility of visually impaired individuals. Developing more effective assistive navigation solutions will be made easier with the aid of this resource. The progression of mobility aids for visually impaired individuals has transitioned from traditional white canes to sophisticated electronic and AI-powered smart canes. This evolution reflects significant technological advancements aimed at enhancing the autonomy and safety of users.

1.3 Organization

The overall structure of this survey and the interrelationships among its various sections are outlined in Fig. 2. This paper will provide a comprehensive and systematic review of evolution, enabling technologies, and the future potential of smart cane systems designed for visually impaired users. This paper uses several abbreviations, which are listed in Table 1. The detailed organization is as follows Table 2:

- **Section I** introduces the background and motivation for this study. It presents global statistics on visual impairment, discusses the mobility challenges faced by visually impaired individuals, and outlines the evolution of assistive navigation technologies. This section also highlights the major limitations in existing systems, such as indoor navigation constraints, sensor limitations, and human-centered challenges. It concludes by presenting the main objectives of this review.
- **Section II** provides a historical overview of smart cane development. It begins with the traditional white cane, followed by early electronic canes, and culminates in the emergence of AI-powered smart canes equipped with sensing, vision, and real-time computation capabilities.
- **Section III** explores the sensor technologies utilized in smart canes, including ultrasonic, infrared, LiDAR, RGB-D, IMU, and GPS sensors. This section also reviews multi-sensor fusion techniques that combine

Table 2 List of abbreviations

Abbreviations	Full form
3D	Three Spatial Dimensions of Width, Height, and Depth
AI	Artificial Intelligence
BLE	Bluetooth Low Energy
CNNs	Convolutional Neural Networks
CPS	Cyber-Physical System
DID	Decentralized Identity
DRL	Deep Reinforcement Learning
DQNs	Deep Q-Networks
DWA	Dynamic Window Approach
ECC	Elliptic Curve Cryptography
ETAs	Electronic Travel Aids
GPS	Global Positioning System
GIS	Geographic Information System
HCI	Human–Computer Interaction
FL	Federated Learning
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IMUs	Inertial Measurement Units
ISO	International Standards Organization
LiDAR	Light Detection and Ranging
MaaS	Mobility-as-a-Service
ML	Machain Learning
NPU s	Neural Processing Units
RGB	Red Green Blue
RL	Reinforcement Learning
RMSE	Root Mean Square Error
PBAC	Policy-Based Access Control
PPO	Proximal Policy Optimization
TTS	Text-To-Speech
URLLC	Ultra-Reliable Low-Latency Communication
UX	User Experience
WCAG	Web Content Accessibility Guidelines
YOYO	You Only Look Once

complementary sensor data to improve obstacle detection and localization accuracy.

- **Section IV** examines the role of SLAM in enhancing smart cane navigation. It discusses various SLAM modalities (e.g., visual, LiDAR-based), integration with Inertial Measurement Units (IMU) and GPS, and real-world applications in indoor and hybrid navigation environments.
- **Section V** focuses on AI and ML techniques applied to assistive navigation. It covers object detection and recognition, path planning algorithms, and adaptive feedback systems such as voice and haptic cues. The section emphasizes the use of deep learning models and reinforcement learning in improving real-time navigation decisions.

- **Section VI** delves into Human–Computer Interaction (HCI) aspects relevant to smart canes. This includes discussions on cognitive load, user experience, multi-modal feedback preferences, ergonomic design, and the integration of smartphones as mobile AI co-processors in navigation systems.
- **Section VII** presents a comparative analysis of existing smart cane technologies. It contrasts various designs in terms of their sensing capabilities, navigation approaches, feedback mechanisms, and limitations, offering a high-level synthesis of current solutions.
- **Section VIII** identifies current challenges and proposes future research directions for the architecture. Topics include the cost-effectiveness of advanced smart canes, user adoption barriers, lack of standardization, and the potential of 5G/6G, edge AI, and personalized navigation systems to address these issues.
- **Section IX** concludes the paper with a summary of major contributions and insights. It emphasizes the importance of integrating SLAM, AI, and sensor fusion to develop effective assistive navigation systems. To achieve smart canes that are affordable, accessible, and able to adapt to dynamic environments, interdisciplinary efforts are required.

2 Evolution of smart cane technology

The journey of assistive navigation technologies began with the traditional white cane, a simple yet effective tool that provides tactile feedback to visually impaired users. Over time, innovations such as the long cane and foldable designs improved portability and ergonomics (Maicha et al. 2025; Nataraj et al. 2024). Even though the white cane remains an important tool, electronic canes have been developed to enhance their functionality by incorporating ultrasonic, infrared, and tactile feedback mechanisms (Sibu et al. 2023). The next significant leap came with electronic travel aids (ETAs), incorporating ultrasonic sensors to detect nearby obstacles and deliver audio or vibratory feedback. Although early ETAs extended detection range and improved safety, their limited intelligence and inability to interpret contextual information restricted their practical utility. These systems often combine obstacle detection, GPS-based routing, and mobile application integration (Ahmad 2024). With the advent of AI, modern smart canes now support simultaneous localization and mapping (SLAM), object recognition, and multimodal feedback, marking a shift from basic assistance to intelligent, personalized mobility support. Most recently, cutting-edge research has explored the integration of Neural Radiance Fields (NeRF) with SLAM to enhance 3D scene understanding (Rabby and Zhang 2024; Rosinol et al. 2022; Wu et al. 2025a, b). NeRF enables photorealistic,

high-fidelity spatial modeling in real time, offering superior depth estimation and object differentiation in cluttered or low-structure indoor environments—factors crucial for blind navigation (Gao et al. 2025). This innovation represents a new frontier in assistive technologies, promising richer scene reconstruction and smarter path planning capabilities.

2.1 Traditional white canes

White canes have historically served as essential aids for people with visual impairments, acting both as a tactile aid for obstacle detection and as social symbols of disability (Díaz-Parra et al. 2023). As evidenced by archaeological finds from biblical times and ancient civilizations, the visually impaired have used sticks and staffs for navigation for a long period of time (Soares et al. 2021). White canes rely on direct contact with obstacles in order to provide users with immediate tactile feedback. They are useful because they provide immediate feedback via the ground and surrounding structures. As a result of this dependence on physical contact, several limitations are imposed. Since the cane's effectiveness in detecting obstacles is limited to its length, typically around one meter, it is not able to detect obstacles at head or torso height. Furthermore, it does not provide a sense of spatial awareness beyond the immediate vicinity of the user (Mai et al. 2023).

2.2 Early electronic canes

Early electronic cane models emerged in the early 2000s, using technologies such as ultrasonic and infrared sensors to overcome traditional canes' inherent limitations. These devices aimed to extend obstacle detection beyond tactile limits and enhance environmental perception.

- Ultrasonic-Based Canes:** Instruments like the Ultra-Cane employ ultrasonic transducers to detect objects up to three meters away. The device emits sound waves via dual sensors and interprets the time-of-flight of the reflected signals to determine object proximity. Tactile feedback is provided via vibration actuators embedded in the handle, with intensity and frequency corresponding to obstacle distance and direction (Nataraj et al. 2024).
- Infrared-Based Canes:** Some early prototypes utilized active infrared emitters and photodiode receivers to detect obstacles within a 3–5 m range. Feedback mechanisms included audio tones and vibrations. Despite their potential, infrared-based systems were prone to performance degradation in bright sunlight and interference from ambient infrared sources, limiting their usability in outdoor environments (Okolo et al. 2024).

While these innovations introduced significant improvements in range and obstacle detection, they also presented new challenges. Ultrasonic sensors suffered from **limited angular resolution** and difficulty detecting soft or narrow obstacles, while infrared systems were susceptible to **environmental interference**. The vibration-based user interfaces also required training and adaptation, presenting a cognitive burden for some users.

2.3 AI-powered smart canes

The latest generation of smart canes incorporates **AI, computer vision**, and **IoT** technologies to deliver dynamic, context-aware navigation support (Maicha et al. 2025; Nokhwal et al. 2023a). These systems typically integrate **GPS, SLAM, depth cameras**, and **ML algorithms** for real-time obstacle classification and environment mapping (Gharghan et al. 2024; Naved et al. 2023).

- i. **Computer Vision and AI Integration:** Modern smart canes, such as the WeWALK Smart Cane (Messaoudi et al. 2022), use AI and computer vision technologies to give users real-time information about their surroundings. The cane integrates a voice-based AI assistant, allowing users to receive contextual information about points of interest, public transportation, and safe walking directions. Vision algorithms, often based on CNNs and object detectors like YOLO, enable the classification of dynamic and static obstacles (Naved et al. 2023).
- ii. **Wearable and IoT-Based Navigation:** Contemporary systems also adopt IoT frameworks, linking the smart cane to mobile applications and cloud services (Veerasamy et al. 2023). Some designs use GPS and cellular networks to transmit the user's location to caregivers, while others incorporate wearable haptic devices for multimodal feedback (Jivrajani et al. 2023). These systems provide real-time, bi-directional communication, enhancing safety, situational awareness, and user confidence (Messaoudi et al. 2022).

AI-powered smart canes thus represent a paradigm shift in assistive technology (Panazan and Dulf 2024). By leveraging ML, edge computing, and sensor fusion, they offer a level of autonomy and environmental awareness previously unattainable. Nonetheless, challenges remain, particularly regarding hardware miniaturization, energy efficiency, user comfort, and privacy concerns arising from collecting sensitive location and visual data.

2.4 Discussion

The evolution of mobility aids—from traditional white canes to AI-powered smart systems—reflects a shift from passive tactile feedback to intelligent, context-aware navigation. While white canes provide immediate, low-tech obstacle detection, they lack anticipatory awareness and pose limitations in complex environments. The first electronic canes used sensors such as ultrasound to extend their detection range, but had problems with resolution, interference, and cognitive load. A recent generation of AI-enabled canes combines computer vision, SLAM, and IoT connectivity to allow users to comprehend semantics, receive dynamic feedback, and monitor their condition in real time. In particular, the integration of NeRF-enhanced SLAM introduces advanced scene reconstruction techniques that offer improved object-level awareness and environmental modeling, key capabilities for dynamic obstacle avoidance and indoor navigation. These advancements can significantly reduce localization drift and enhance confidence in complex, unfamiliar spaces. Smart canes with smart technology are increasingly viewed as multi-functional aids—empowering not only navigation, but also gaming, social networking, and real-time health monitoring (Messaoudi and Dhiaeddine 2024). In spite of the advances, some challenges persist, such as those related to processing demands, ergonomics, energy efficiency, and data privacy. According to Table 3, key smart cane technologies are compared, highlighting their capabilities and limitations.

3 Sensor technologies in smart canes

Smart canes rely on various sensor technologies to perceive and interpret the surrounding environment. Ultrasonic sensors are widely used for their affordability and simplicity in detecting obstacles within a short range. Infrared sensors, while similar, are more sensitive to ambient light conditions and are typically used for surface or step detection. LiDAR and RGB-D cameras offer high-resolution depth perception and are suitable for advanced applications such as 3D mapping and SLAM. Inertial Measurement Units (IMUs) assist with orientation and motion tracking, while GPS modules support outdoor localization. Many systems also incorporate environmental sensors for light, humidity, or temperature, enhancing situational awareness. The effectiveness of these sensors depends on their integration, power consumption, and computational requirements, which must be optimized for real-time processing in a wearable device. Table 3 presents a comparative analysis of the performance of these sensors. Choosing appropriate sensor technology for smart canes requires careful consideration of their functionalities, benefits, and limitations.

Table 3 Comparative analysis of sensor technologies in smart canes

Sensor Type	Functionality	Advantages	Disadvantages	Typical Applications
Ultrasonic Sensors	Emit high-frequency sound waves to detect obstacles by measuring the echo return time	Cost-effective- Simple implementation- Real-time obstacle detection	Limited accuracy in complex environments- Short detection range- Susceptible to interference from environmental noise	Obstacle detection in smart canes, e.g., SmartCane
Infrared Sensors	Use infrared light to detect objects and measure distance	Accurate depth sensing- Effective in various lighting conditions	High cost- Performance is affected by environmental factors like rain or fog	Object detection and distance measurement in electronic travel aids
LiDAR Sensors	Utilize laser pulses to create detailed 3D maps of the environment	High-resolution environmental mapping- Effective in various lighting conditions	High cost- Sensitivity to adverse weather conditions- Larger size and power consumption	Environmental mapping and obstacle recognition in advanced navigation systems
RGB-D Cameras	Capture color (RGB) and depth (D) information for 3D perception	Facilitates computer vision-based navigation- Enables AI-powered object recognition	Performance can be affected by lighting conditions- Requires significant processing power	Object recognition and navigation assistance in smart canes
IMUs	Measure acceleration and angular velocity to track movement and orientation	Compact and low power consumption- Provides real-time motion data	Accumulation of errors over time (drift)- Requires calibration	Motion tracking and localization in assistive devices
GPS and GIS	Provide global positioning data and geographic information	Enables outdoor navigation and wayfinding- Wide availability	Reduced accuracy in urban canyons or indoors- Dependence on satellite visibility- signal blocking	Urban navigation and route planning

3.1 Ultrasonic sensors

An ultrasonic sensor emits high-frequency sound waves and measures the echo return time in order to detect obstacles. They are cost-effective, simple, and respond in real-time, making them suitable for basic navigation tasks (Meshram et al. 2019). However, ambient noise causes them to be less accurate in complex environments, have a limited range (typically only four meters), and have a poor angular resolution. For instance, the Smart Cane (IIT Delhi) employs ultrasonic sensors to detect obstacles and alert users through vibrations.

3.2 Infrared sensors

Sensors measure distance using reflected IR light, offering moderate accuracy and resistance to ambient lighting, but they are often costly and can be affected by environmental factors such as rain or fog (Zafar et al. 2022). It is good for short-range detection (up to 2 m), affordable, and compact. **It can be used in** object proximity detection for low-cost electronic travel aids (ETAs).

3.3 LiDAR

Infrared (IR) sensors measure distance using reflected IR light and offer moderate accuracy with some resistance to ambient lighting conditions. There are two main types: passive IR sensors, which are compact, affordable, and commonly used for short-range proximity detection (typically up to 2 m), and active or high-resolution IR systems, which may incur higher costs due to their precision and enhanced capabilities. While IR sensors are well-suited for low-cost electronic travel aids (ETAs), they can be affected by environmental factors such as rain, fog, or reflective surfaces, which may degrade performance (Zafar et al. 2024). Despite their effectiveness in diverse lighting conditions, their high cost and sensitivity to adverse weather conditions limit their widespread adoption (Zafar et al. 2024). An example of LiDAR application is seen in smart canes that integrate 2D LiDAR and RGB-D cameras for navigation and obstacle recognition (Feng et al. 2025). Selecting the appropriate IR sensor thus involves balancing cost, range, and reliability based on the intended use scenario.

3.4 RGB-D cameras

Enhance computer vision-based navigation by capturing both color and depth information, enabling AI-powered object recognition (Mai et al. 2023). Nevertheless, their performance can be compromised under varying lighting conditions, and they require substantial processing power: real-time object detection, semantic SLAM (e.g., YOLO-based detection).

3.5 Inertial measurement units (IMUs)

IMUs are valuable for tracking movement and orientation, offering compact size and low power consumption (Feng et al. 2025). IMUs are compact, low-power, and enable dead-reckoning in GPS-denied environments. However, they are prone to error accumulation over time (drift) and necessitate regular calibration. IMUs are integral in motion tracking and localization within assistive devices (Zafar et al. 2024).

3.6 GPS and GIS

GPS provides real-time global positioning, while Geographic Information System (GIS) adds contextual geographic data for intelligent route planning (Gad et al. 2023; Feng et al. 2025). While they enable effective wayfinding, their accuracy can diminish in urban canyons or indoor environments due to signal obstruction and reliance on satellite visibility (Abujayyab et al. 2023).

3.7 Discussion

Sensor fusion, not individual superiority, is central to advancing smart cane design. Each sensing modality—ultrasonic, infrared, LiDAR, RGB-D, IMUs, and GPS—offers distinct strengths but also critical limitations. For instance, ultrasonic sensors are cost-effective for short-range detection but struggle in noisy environments, while LiDAR offers high spatial resolution at the cost of size, power, and weather sensitivity. Sensors such as RGB-D and infrared enhance depth perception and the ability to recognize objects, but they are also affected by lighting conditions and the environment. Motion tracking and outdoor localization can be improved using IMUs and GPS, but these technologies are hampered by drift and signal loss. Based on this analysis, it is clear that synergistic sensor fusion strategies are required to overcome individual weaknesses and to improve situational awareness. The development of hybrid architectures – for example, combining ultrasonics with RGB-D for low-latency detection and semantic perception, or GPS with IMUs for robust localization – allows the development of a more reliable and context-aware navigation system. There are, however, new challenges associated with this integration, such as increased power requirements, processing latency, and ergonomic concerns. Designing future systems should prioritize adaptive sensor management, low-latency pipelines, and user-focused configurations tailored to individual mobility patterns and environments. An intelligent orchestration of senses, perceptions, and real-time decision-making will be essential to the future evolution of smart canes from basic aids to autonomous, context-aware mobility partners.

4 Advanced navigation technologies in smart canes: the synergistic role of SLAM and AI

The field of assistive navigation for visually impaired individuals has evolved significantly over the past few years, particularly with the integration of SLAM (Bamdad et al. 2024) and AI (Walle et al. 2022). These technologies serve complementary roles: SLAM enables spatial awareness and localization, while AI empowers perception, decision-making, and adaptive behavior. Their convergence, supported by advances in embedded computing and sensor fusion, forms the backbone of intelligent navigation systems in smart canes.

4.1 Historical evolution of SLAM in assistive navigation

The application of SLAM in mobile robotics dates back to the mid-1990s, primarily used for autonomous indoor mapping using Kalman filters (Ahmad and Alkhamash 2024) and later expanded to particle filters and probabilistic frameworks (Walle et al. 2022). However, SLAM's transition to assistive technologies for the visually impaired accelerated only after 2010, driven by advances in sensor miniaturization and the availability of low-power, embedded computing platforms such as NVIDIA Jetson Nano and Raspberry Pi. This evolution is illustrated in Fig. 3, tracing SLAM's journey from basic visual methods to semantically rich, hybrid systems.

- i. **Early Approaches (2012–2016):** Initial systems relied on monocular and stereo vision with SLAM variants such as ORB-SLAM (feature-based) and LSD-SLAM (direct method). These algorithms enabled sparse and semi-dense mapping but suffered in textureless or low-light environments, limiting their real-world applicability (Bamdad et al. 2024). Visual SLAM (V-SLAM), although effective in controlled settings, was vulnerable to motion blur and occlusions.
- ii. **Middle-Phase (2017–2020):** The integration of 2D/3D LiDAR sensors led to the rise of Laser SLAM, offering denser and more accurate environmental representations. These models were typically fused with IMU and RGB-D data to improve robustness against sensor noise and dynamic environments (Jain and Patel 2023; Slade et al. 2021). Figure 1 illustrates such a system architecture incorporating LiDAR and inertial measurements into a smart cane design.
- iii. **Contemporary Hybrid Systems (2021–2024):** Recent advancements introduced hybrid SLAM models that

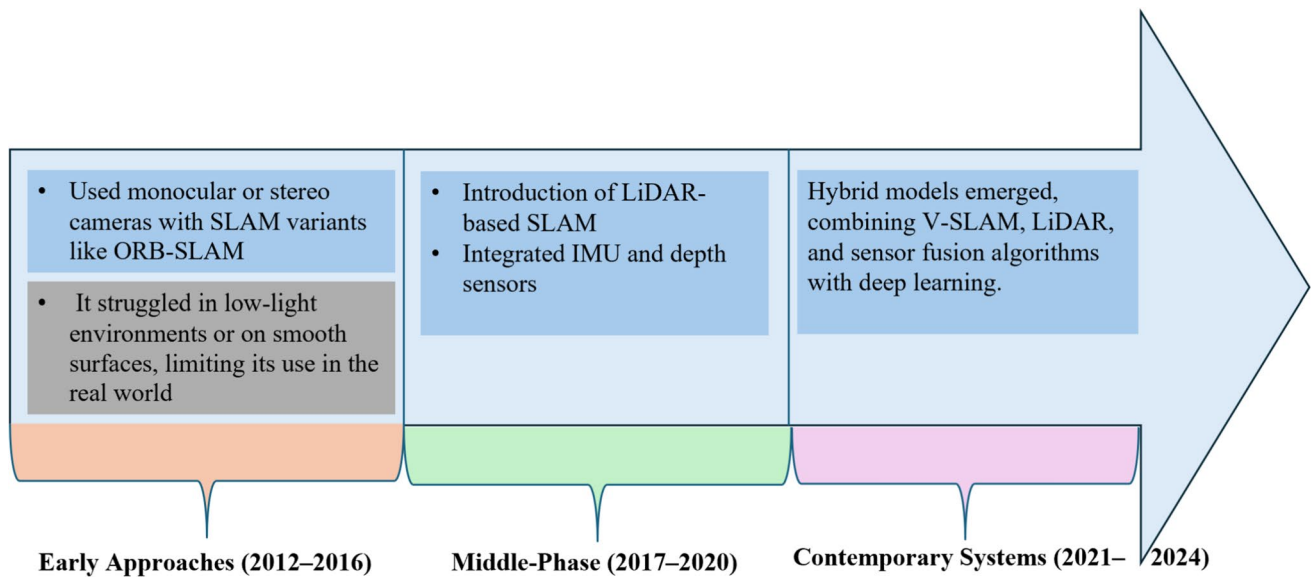


Fig. 3 Evolution of SLAM technologies for assistive navigation: from monocular approaches to deep learning-enabled hybrid models (2012–2024)

fuse LiDAR, V-SLAM, and deep learning (Beingolea et al. 2021). These deep semantic SLAM models combine RGB-D or LiDAR-based geometric mapping with CNN- or Transformer-based object recognition modules to construct semantic maps, where objects and regions are labeled by class, utility, and risk. Key innovations include:

- **Dynamic SLAM Handling:** Traditional SLAM systems treated moving objects as noise. Contemporary hybrid SLAM systems—such as DS-SLAM and DynaSLAM—now explicitly detect, track, and filter out dynamic elements using optical flow and learning-based foreground segmentation, ensuring map consistency in crowded or changing scenes.
- **Neural Radiance Fields (NeRFs):** Though originally developed for 3D scene reconstruction, NeRF-SLAM variants now allow smart canes to perform continuous scene representation with high fidelity, even under variable lighting or partial occlusion. NeRFs support predictive modeling, enabling canes to infer the appearance or geometry of partially visible objects.
- **Online Learning and Adaptation:** Edge-deployable models using incremental learning (e.g., using life-long SLAM or online updating CNNs) enable systems to adapt to user-specific contexts, improving accuracy over time. This is essential for accommodating different walking speeds, mobility patterns, and frequently visited environments.

- **Multi-modal Fusion Frameworks:** Modern systems use graph optimization with sensor nodes representing LiDAR, IMU, camera, and GPS data streams. Tools like GTSAM and Ceres Solver enable real-time, high-precision map optimization, incorporating temporal and spatial constraints for robust localization across diverse terrain.

Case Study: Mai et al. (2024) developed a smart cane system integrating 2D LiDAR, RGB-D camera, GPS, and IMU to assist visually impaired users by enabling SLAM-based navigation and real-time object recognition using an improved YOLOv5 algorithm. Tested in both indoor and outdoor environments, the system achieved high accuracy in obstacle detection and mapping, with recognition speeds of 25–31 FPS and support for 86 object types, offering autonomous guidance akin to a "guide dog" (Mai et al. 2024).

4.2 AI-enhanced SLAM for object recognition and scene understanding

While SLAM handles spatial mapping, it lacks semantic awareness. AI, especially deep learning models, fills this gap by enabling the recognition of objects, people, and terrain types (Nagarajan and M P, 2023). Integration of AI within SLAM pipelines allows the smart cane not only to detect where obstacles are but also what they are.

- i. **YOLO** and its variants (v4, v5, v7) are widely used for real-time object detection due to their balance between accuracy and speed (Mai et al. 2024; Theodorou et al. 2023).
- ii. **CNNs** and **Region-based CNNs (R-CNNs)** are often used for more detailed classification tasks (Walle et al. 2022).
- iii. Integration into SLAM is typically done via semantic SLAM, where bounding box data from detectors is associated with 3D points from the SLAM map to create annotated environmental models (Bamdad et al. 2024).

Recent Advancement: Theodorou et al. (2023) proposed an AI-SLAM framework integrating YOLOv7 with V-SLAM, achieving mAP of 94.3% on real-time indoor obstacle classification and mapping (Theodorou et al. 2023). A noteworthy recent advancement in this area is the integration of NeRF with SLAM systems to enhance 3D scene representation and semantic comprehension. NeRF leverages volumetric rendering and neural networks to reconstruct high-fidelity scenes from sparse visual inputs, enabling robust occlusion handling and fine-grained object boundaries. In the context of smart canes, NeRF-SLAM can significantly improve scene perception in cluttered or dynamic indoor environments, offering more reliable spatial reasoning for path planning and obstacle avoidance. When combined with AI-based semantic segmentation, these systems allow for more detailed and context-aware environmental models, pushing the boundary of what assistive navigation systems can achieve in real-time.

4.3 Sensor fusion architecture for smart canes

Modern smart canes adopt multi-modal sensor fusion strategies to overcome individual sensors' limitations (Gharghan et al. 2023). No single sensor type offers a complete solution, as explained in Section 3. Sensor fusion techniques integrate data from multiple modalities to compensate for individual shortcomings, as shown in Fig. 4. In the sensor input layer, data is collected from multiple sensors, including ultrasonic sensors, RGB-D cameras, LiDAR units, IMUs, and GPS modules. Each sensor contributes a unique type of information—ultrasonics provide simple distance measurements, RGB-D cameras offer depth perception, LiDAR delivers precise 3D mapping, IMUs track motion and orientation, and GPS enables outdoor positioning. Once the raw data is gathered, it is passed to the fusion layer, where two primary strategies are employed to integrate this multimodal information. The first approach involves using a Kalman Filter. This classical statistical method merges noisy measurements such as GPS signals and IMU readings to produce a more reliable estimate of the user's position and movement. The second approach leverages deep fusion techniques, where AI models like CNN-LSTM networks are applied to learn complex temporal and spatial relationships within the data streams, leading to improved obstacle detection and localization (Ben Atitallah et al. 2023). To further strengthen the architecture, recent works in autonomous driving and IoT-based motion systems offer transferable insights. For example, dynamic multi-sensor fusion frameworks with adaptive spatiotemporal optimization as discussed in (Li et al. 2025) can be leveraged to enhance motion understanding in assistive devices. Similarly,

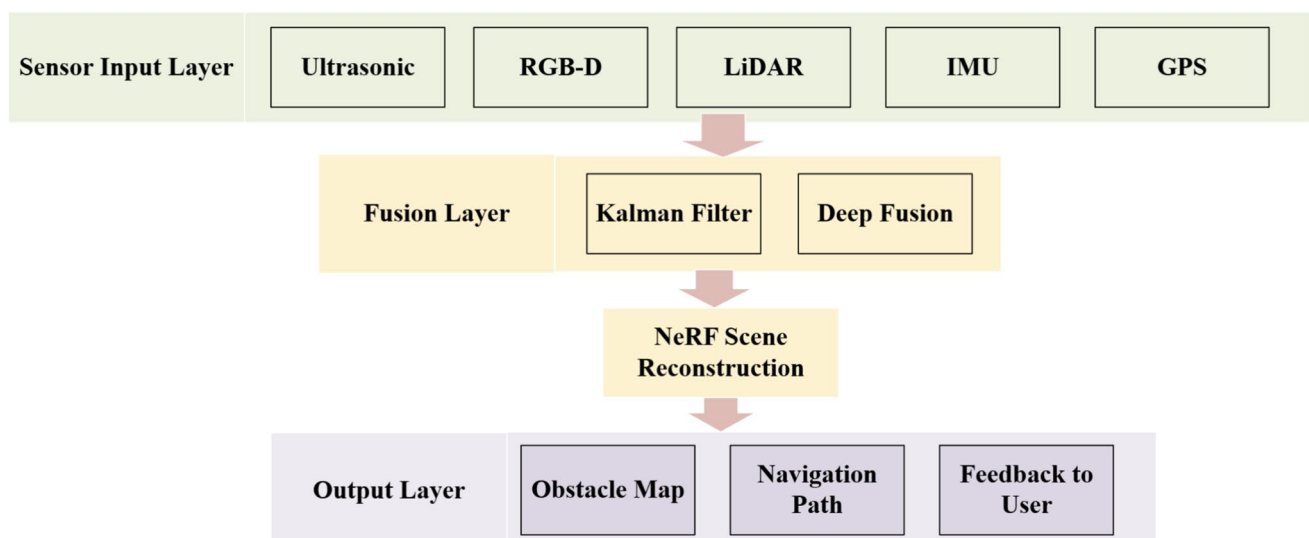


Fig. 4 Sensor fusion architecture for smart canes: Integration of multimodal sensors via fusion algorithms to generate obstacle maps, navigation paths, and multimodal user feedback

cooperative perception mechanisms used in autonomous vehicles (Xiang et al. 2023) present an opportunity for inter-device collaboration in future smart cane networks. Furthermore, diffusion models have emerged for robust multi-sensor fusion in 3D object detection and BEV segmentation, offering a promising research direction to handle uncertainty and sparsity in real-world sensor data. In the output layer, the fused data is used to generate actionable insights such as constructing obstacle maps, determining safe navigation paths, and delivering real-time, context-aware feedback through audio, haptic, or visual channels. These processes not only enhance the safety and autonomy of the user but also provide the foundation for intelligent, adaptive mobility systems.

In modern smart cane systems, sensor fusion strategies are further categorized based on the data integration stage. These strategies define how raw or processed sensor information is combined to enhance environmental understanding and navigation accuracy. Three primary fusion approaches are commonly employed to optimize system performance: early, late, and AI-based. Early Fusion combines raw sensor data before interpretation (e.g., LiDAR + RGB-D for SLAM). Late Fusion merges processed outputs (e.g., IMU trajectory + GPS location). AI-Based Fusion: Deep learning models (e.g., CNN-LSTM) for temporal and spatial data fusion (Hocaoglu and Benli 2025). Sensor fusion increases robustness and allows cross-validation of sensor data, which is crucial for safety-critical navigation. Finally, the processed and fused information reaches the output layer, where it generates actionable results. This includes constructing an obstacle map that visually represents detected barriers in the user's environment, dynamically planning a navigation path to guide safe movement, and delivering real-time feedback to the user through audio commands, haptic signals, or visual cues via connected smart devices.

In recent developments, NeRF has been explored as a post-fusion enhancement method, particularly in systems leveraging RGB-D and LiDAR inputs. NeRF utilizes fused visual and spatial data to reconstruct high-resolution 3D scenes, improving scene continuity, depth estimation, and occlusion handling. When incorporated into smart cane architectures, NeRF can enable richer environmental modeling and support advanced SLAM features such as semantic layering and photorealistic navigation cues. Though still in early stages of integration for wearable assistive systems, NeRF-SLAM models represent a promising direction for future multimodal navigation frameworks.

4.4 Path planning algorithms in smart canes

SLAM provides the spatial layout; AI enables the planning of optimal paths through this space. Traditionally, path planning used deterministic graph-based algorithms:

A* and Dijkstra's algorithm are effective in static, grid-based environments but are limited in dynamic contexts (Kaiyue et al. 2025; De Mel et al. 2024). Reinforcement Learning (RL) models, particularly Deep Q-Networks (DQNs) and Proximal Policy Optimization (PPO), have emerged as state-of-the-art for real-time, adaptive navigation (Tezerjani et al. 2024; Wen et al. 2020). RL-based agents learn to navigate unfamiliar terrains by reward-based interactions, enabling human-like decision-making in complex scenarios. Empirical result by Patel et al. (2020a) presents a Deep Reinforcement Learning (DRL) policy that combines the Dynamic Window Approach (DWA) with DRL for navigating between mobile obstacles (Patel et al. 2020a). Compared with state-of-the-art collision avoidance methods, the proposed method achieved a 33% higher success rate and a 61% lower rate of violations of dynamics constraints (Patel et al. 2020b). Recent developments have further pushed the frontier by enabling collaborative and privacy-preserving path planning strategies: Wu et al. (2025a, b) proposed a Cooperative Deep Reinforcement Learning (CDRL) model designed for multi-agent environments. This technique facilitates knowledge sharing between agents, allowing for distributed smart cane systems to coordinate navigation and obstacle avoidance, especially in crowded environments (Wu et al. 2025a, b). Wu et al. (2024) introduced a Federated Learning-Based Path Planning Framework that maintains user privacy while training navigation models across decentralized devices. Such an approach is highly relevant for smart cane systems that process sensitive data, as it eliminates the need for central data aggregation, ensuring user confidentiality without compromising model accuracy (Wu et al. 2024).

4.5 Multimodal feedback: bridging AI and human sensory needs

An intelligent navigation system must translate machine-generated decisions into intuitive cues for the user. Two dominant modalities have emerged: Voice Feedback: Text-To-Speech (TTS) engines, often in local languages, provide real-time directional or object-related information (Niveditha et al. 2024). Haptic Feedback: Tactile actuators (e.g., vibration motors) provide directional signals, proximity alerts, or obstacle classification in a non-intrusive manner (Niveditha et al. 2024). These feedback systems are now often context-aware, adjusting the mode or intensity based on environmental complexity or user preferences. *Example:* A 2023 design by Honda et al. Implemented spatialized audio coupled with haptic vibration to alert users of fast-moving obstacles, improving reaction time by 32% (Honda et al. 2023).

4.6 Discussion

Integrating SLAM and AI into smart canes marks a transformative leap in assistive navigation for the visually impaired. SLAM, once a robotic mapping tool, now enables real-time localization via visual, LiDAR, and inertial sensors, as shown in Fig. 5. AI, evolving from basic object recognition to contextual understanding and decision-making, adds semantic depth to SLAM maps—classifying objects, predicting motion, and prioritizing obstacles. Modern hybrid SLAM systems, enhanced by deep learning, offer robust indoor-outdoor performance. Sensor fusion via Kalman filters or CNN-LSTM models, as shown in Fig. 6, ensures resilience against noise, occlusion, and signal loss. Meanwhile, navigation has shifted from static A* paths to RL for adaptive, human-like route planning. Crucially, intuitive voice and haptic feedback bridge the system-user gap, optimizing safety and reducing cognitive load. SLAM and AI are convergent technologies that transform smart canes from reactive tools into proactive companions, enhancing autonomy, safety, and dignity. There are several challenges that remain, such as cost, size, energy efficiency, privacy, and individualization of the models. The direction of the future is collaborative navigation, continuous learning, and embedded artificial intelligence, paving the way for next-generation mobility devices.

5 Human-centered interfaces as enablers of intelligent feedback in smart canes

While advances in SLAM, AI, and sensor fusion form the backbone of modern smart cane systems, their real-world efficacy depends on how effectively these technologies interface with human users. Rather than treating Human–Computer Interaction (HCI) as an isolated focus, this section frames it as a critical enabler of intelligent feedback, complementing the perception and decision-making layers within smart cane architecture. Smart cane effectiveness is not solely determined by obstacle detection or localization accuracy, but also by the clarity, timing, and intuitiveness of the feedback delivered to the user. For instance, multisensory cues—such as haptic pulses, voice instructions, or auditory tones—must be designed to reduce cognitive load and promote rapid comprehension in diverse environments (Soares et al. 2021). This is a framework for analyzing how users interact with smart cane systems. Poorly timed or ambiguous feedback can erode trust in the system and hinder adoption, even if the underlying AI models perform well. Human-centered design principles ensure that assistive navigation remains cognitively manageable, ergonomically feasible, and socially acceptable. Rather than a standalone topic, HCI is interwoven into the system’s architecture as the final translation layer between

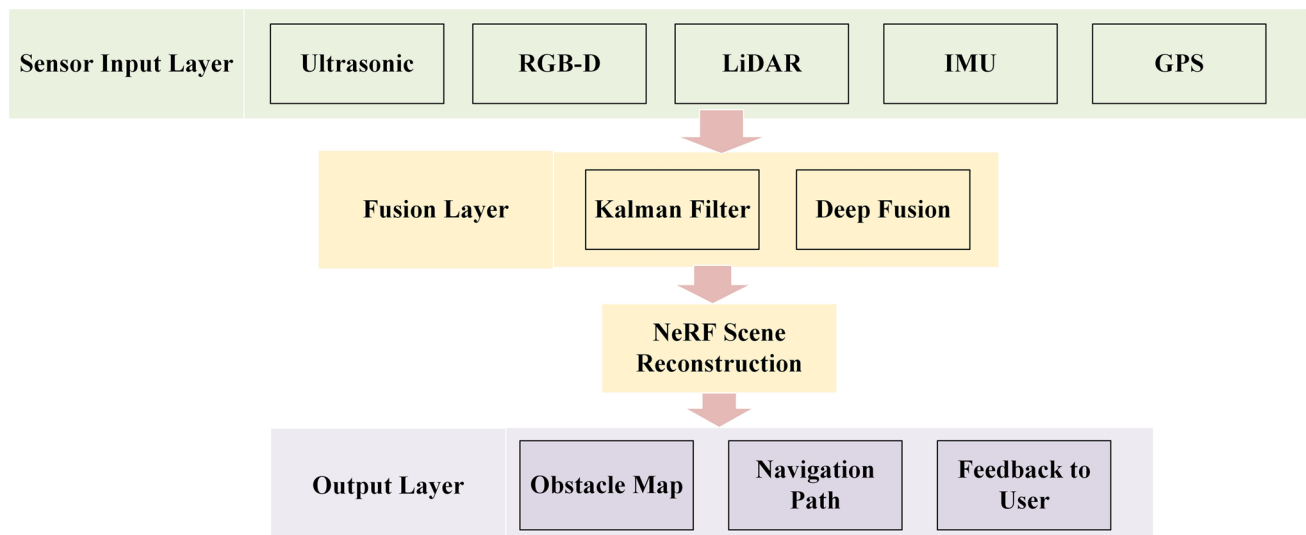
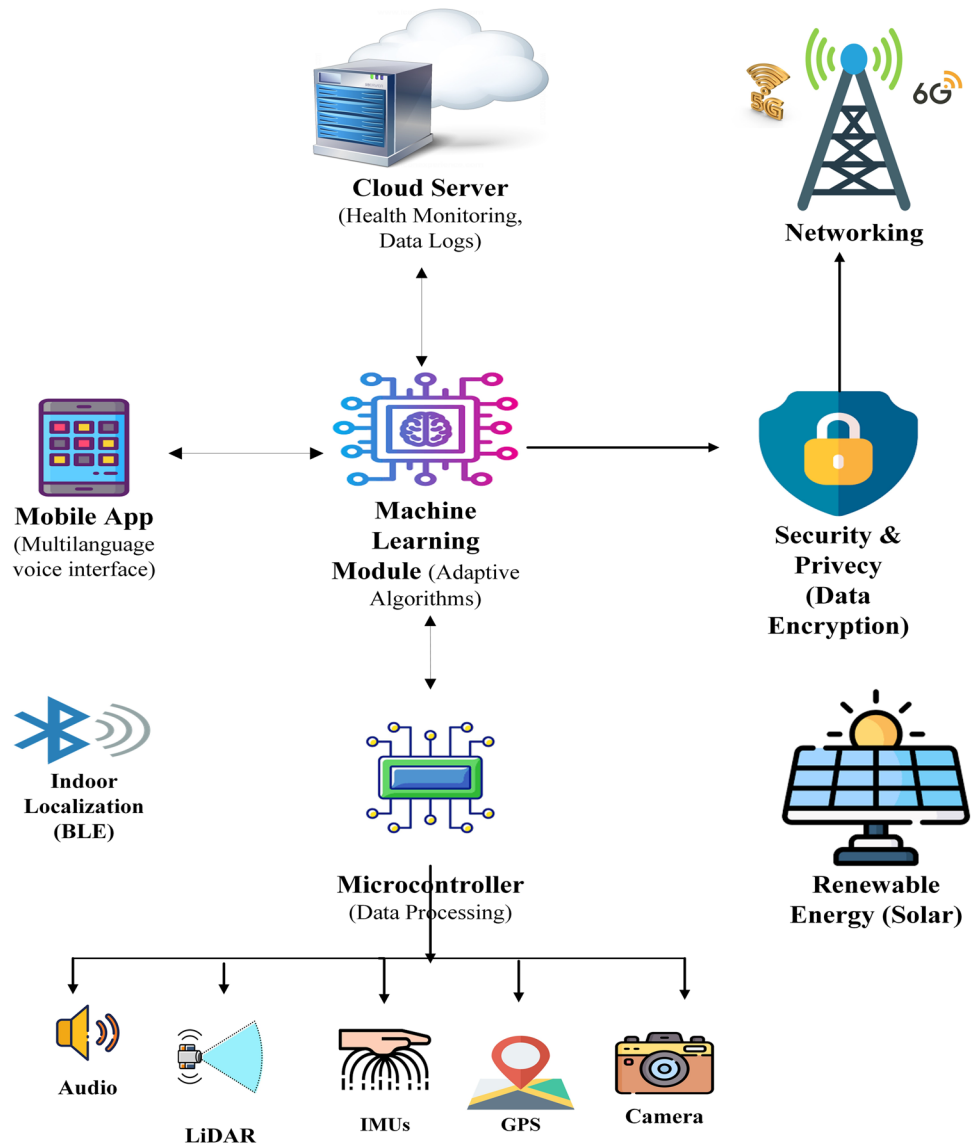


Fig. 5 Multi-layered HCI framework for smart cane systems

Fig. 6 Proposed future smart cane framework for the visually impaired: integrating machine learning and IoT technologies



perception (sensors), decision-making (AI), and actionable user guidance (feedback). Prioritizing accessible interfaces, lightweight form factors, and adaptive personalization—such as multilingual voice prompts or customizable feedback modes—can significantly increase usability and long-term user acceptance. In this context, HCI functions as the conduit through which AI-driven decisions and sensor-derived insights are rendered meaningful and actionable. Future research should continue to embed HCI considerations into the technical design cycle, ensuring that smart canes do not just perceive and decide, but also communicate effectively, inclusively, and intuitively. To ensure accessibility, developers are encouraged to follow established design guidelines, such as those outlined in WCAG 2.1, and to validate systems through participatory testing with target users.

5.1 User experience (UX) and accessibility

- i. **Cognitive Load in Smart Cane Usage:** A major concern in HCI for assistive devices is the reduction of cognitive load. As visually impaired users rely heavily on their auditory and tactile senses, a sudden influx of information can result in sensory overload, increased stress, and a loss of confidence in the device. Recent studies (Honda et al. 2023; Skovfoged et al. 2022a; Zhang et al. 2024) highlight that multimodal feedback can confuse rather than aid the user if not hierarchically structured. Smart cane systems must prioritize essential alerts, such as immediate obstacles or directional changes, while deferring or suppressing secondary cues (e.g., environmental descriptions). Advanced UX models suggest implementing adaptive information hierar-

chies, where the system learns the user's behavior over time and tailors the feedback level dynamically based on context (e.g., crowded vs. open space).

- ii. **Preference for Multimodal Feedback:** Extensive user evaluations show a clear preference for multimodal feedback, combining (Hocaoglu and Benli 2025; Theodorou et al. 2023): Auditory Cues: For directional instructions and object classification (e.g., "stairs ahead" or "left turn in 3 m"). Haptic Signals: For silent, non-distracting alerts, especially in public or noisy environments. Optional Visual Interfaces (for users with partial vision): Visual overlays on smartphone apps or AR glasses can offer optional supplementary cues. *Case Study:* A 2023 user trial by Honda et al. This study investigates how combining vibration and electrical stimulation affects the latency of kinesthetic illusions, which are sensations of movement without actual motion. While it doesn't directly address task completion time or NASA-TLX scores, it highlights the potential of multimodal feedback in enhancing user perception and response (Honda et al. 2023). The NASA task load index (NASA-TLX) is a tool for measuring and conducting a subjective mental workload assessment.

5.2 Ergonomics and Portability

- i. **Weight & Size Considerations in Wearable Navigation Systems:** Despite advancements in miniaturization, one persistent challenge in smart cane development is balancing functionality with portability. Integrating SLAM, LiDAR, AI modules, depth cameras, IMUs, microcontrollers, GPS, and batteries can result in a bulky and cumbersome device, which may hinder usability for visually impaired users. Therefore, the design must prioritize technological integration, physical comfort, and portability. Key considerations include (Kan et al. 2025; A. Khan et al. 2018):
 - **Weight Distribution:** Evenly distributing the system's weight along the cane shaft helps prevent wrist fatigue and improves overall handling.
 - **Battery Placement:** Using counterbalancing techniques or relocating batteries to a waist-worn or belt-mounted pack can reduce strain on the cane itself.
 - **Form Factor:** Foldable or modular designs enhance portability and storage convenience without compromising structural integrity.
 - **Compact Integration:** Selecting lightweight, multifunctional components and embedding them into a unified system helps minimize bulk.

- **Ergonomics and Usability:** The cane should remain comfortable for prolonged use, with intuitive access to key functions.
- **Power Efficiency:** Utilizing low-power AI processors and optimizing energy consumption allows smaller batteries and longer operational time.

Together, these design principles ensure that smart canes remain both technologically powerful and physically practical for everyday use. According to Kan et al. (2025), users showed that eLabrador is a wearable navigation system designed to assist visually impaired users in outdoor environments by combining GPS, public maps, and visual perception via a head-mounted RGB-D camera. It generates a 3D semantic map for local guidance and relays navigation through audio-haptic feedback. Real-world trials showed effective guidance, improved environmental awareness, and user-reported safety and usability (Kan et al. 2025).

- ii. **Wearability and Hands-Free Operation:** To enhance mobility and reduce the need for constant hand engagement, emerging assistive navigation systems are exploring wearable designs that integrate critical sensing and feedback components into the user's body gear. These include (Bouteraa 2023; Kan et al. 2025), 1) Belt-mounted LiDAR units for continuous obstacle detection. 2) Chest-mounted haptic feedback systems to convey directional cues through vibrations. 3) Smart glasses with V-SLAM capabilities for visual localization and real-time mapping. While these innovations support greater freedom of movement and spatial awareness, successful deployment hinges on thoughtful ergonomic and engineering considerations. Devices must maintain low thermal output to prevent user discomfort, exhibit minimal obtrusiveness to avoid interfering with natural movements or daily tasks, and ensure consistent and reliable skin contact for effective haptic communication (Bouteraa 2023). Durability, discreet form factors, and intuitive user interaction are essential to facilitate adoption and sustained use in everyday environments.

5.3 Smartphone integration in assistive navigation systems

Smartphones as AI Co-Processors for SLAM-Based Smart Canes: The integration of smartphones as external cognitive processing units represents a pivotal advancement in the design of smart cane systems for visually impaired users (Ullah et al. 2022). By offloading computationally intensive tasks, such as object detection, real-time localization, and natural language interaction, to smartphones, researchers and engineers can significantly reduce the onboard

hardware complexity and energy requirements of assistive devices (Kuriakose et al. 2023; Wang et al. 2024). Modern smartphones are increasingly equipped with dedicated Neural Processing Units (NPUs) (Nokhwal et al. 2023b) or AI accelerators (Subhan et al. 2024), capable of delivering high-throughput, low-latency execution of ML algorithms on-device. These capabilities position smartphones as ideal AI co-processors for SLAM-based navigation systems. Specifically, they support:

Real-time object recognition using optimized deep learning models (e.g., YOLOv5, MobileNet-SSD), enabling timely detection of dynamic and static obstacles (Islam et al. 2023).

Visual-inertial SLAM processing through advanced mobile Software Development Kits (SDKs) such as Google ARCore and Apple ARKit allows robust environmental mapping and localization with minimal additional hardware.

Natural interaction support via on-device speech recognition, voice command processing, and TTS feedback mechanisms, promoting a multimodal and accessible user interface.

This architecture transforms the smartphone from a peripheral device into a cognitive hub, centralizing computation, reducing the smart cane's weight, and enabling more ergonomic and energy-efficient designs (Kuriakose et al. 2023). Empirical Example: In a study by Hu et al. (2024), SEESys is a real-time SLAM pose error estimation system that uses a low-overhead runtime monitor and a multi-modality deep learning model (DeepSEE) to deliver accurate, low-latency error predictions. It achieves 15.8% better Root Mean Square Error (RMSE) than baselines and improves pose tracking accuracy by 25% through real-time audio feedback. The smartphone-enabled design supports portability and affordability and leverages widely adopted consumer technologies, promoting scalability and user familiarity (Kuriakose et al. 2023). As AI models continue to be optimized for edge devices and as mobile NPUs grow more capable, the role of smartphones as context-aware assistive intelligence agents is likely to expand, offering a promising direction for the next generation of wearable and mobile navigation systems for individuals with visual impairments.

- ii. **Mobile Application-Based Navigation Support for Visually Impaired Users:** Mobile applications have emerged as an essential interface layer for smart cane systems, enhancing both usability and contextual adaptability through intuitive human–computer interaction (HCI) principles. These apps bridge the gap between complex sensor processing and user-centered accessibility by offering personalized and dynamic navigation assistance. Key functionalities include

(Feng et al. 2025; Jain and Patel 2023; Kaiyue et al. 2025):

- iii. **Voice-based route planning and destination input**, enabling hands-free operation and minimizing cognitive load during navigation.
- iv. **Adaptive feedback customization**, allowing users to modify haptic intensity or audio volume in response to situational demands (e.g., muting voice feedback in libraries or increasing vibration in noisy outdoor environments).
- v. **Real-time geolocation tracking** enables family members or caregivers to monitor the user's position and receive emergency alerts, enhancing safety and trust in outdoor mobility scenarios.
- vi. **Crowdsourced environmental annotations**, whereby users can report and label hazardous areas, such as construction zones, uneven pavements, or potholes, enrich the shared navigation map and benefit the broader visually impaired community.

Modern mobile platforms also enable learning-based personalization through lightweight meta-learning frameworks, which continuously adapt app behavior based on user proficiency, movement patterns, and environmental preferences (Wang et al. 2024). This facilitates a more intelligent and responsive system that evolves in parallel with the user's mobility skills. Many current applications are now leveraging open geospatial platforms like OpenStreetMap, Mapillary, and Google Maps APIs, enhanced with computer vision modules for real-time semantic scene understanding. These systems assist in navigating semi-structured environments such as parks, malls, or university campuses, where traditional GPS signals may be degraded or absent.

Integrating human-centered design principles into these apps ensures that technological sophistication aligns with user expectations and lived experiences (Ullah et al. 2022; Wang et al. 2024). Effective mobile interfaces should emphasize: 1) Minimalist interaction design to reduce cognitive and operational burden. 2) Multi-modal feedback (audio, haptic, visual for low-vision users) to support diverse sensory preferences. 3) Cross-platform compatibility to ensure accessibility across different device ecosystems (Android, iOS) (Khan et al. 2021a, b). Moreover, co-design methodologies—involving visually impaired users at all stages of development—are essential to create inclusive, culturally adaptable, and acceptable technologies. Such collaborative approaches validate usability in real-world contexts and strengthen long-term adoption and trust. Ultimately, mobile app integration elevates smart canes from passive navigational aids to interactive, adaptive assistive ecosystems capable of intelligent interaction, user learning, and community collaboration. As shown in Table 4, comparing key HCI

Table 4 HCI-centric design evaluation of recent smart cane systems (2021–2024)

Study/Device	Feedback Modalities	Ergonomics	Smartphone Integration	User-Centric Findings
Shin et al. (2023)	Audio + Haptic	Modular, 580g	Basic Bluetooth pairing	90% of users reported improved situational awareness
(Honda et al. 2023)	Spatial Audio + Haptic	Wearable belt (LiDAR) + cane	Real-time route updates via app	24% lower mental load (NASA-TLX)
(Hu et al. 2024)	Audio only	Lightweight	Lightweight RTS monitor operates concurrently with a SLAM system on a mobile device	15.8% better RMSE; audio feedback improved pose tracking by 25%; 37.3 ms end-to-end latency
(Díaz-Parra et al. 2023)	Audio + Visual (for low vision)	Foldable cane, adjustable grip	Mobile app with customizable UI	High user satisfaction for customization features
(Patel et al. 2020b)	Haptic-only	Chest-mounted device (hands-free)	None	Lower distraction but high learning curve

design aspects in recent smart cane systems (2021–2024) based on UX, ergonomics, and smartphone integration.

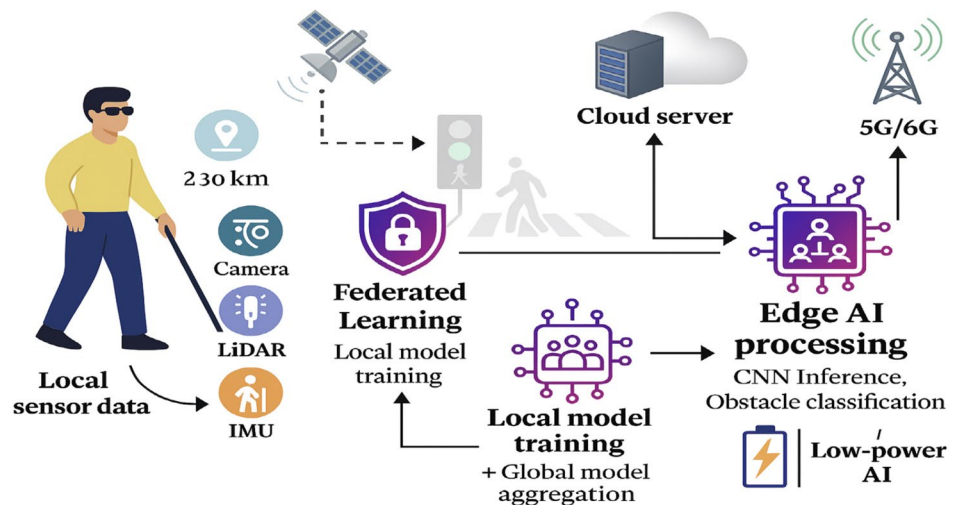
Table 4 compares five recent smart cane systems (2021–2024), focusing on three foundational pillars of HCI: multimodal feedback, ergonomic design, and smartphone integration. This HCI-centric framing offers critical insight into how assistive mobility systems evolve toward more user-aligned, context-aware, and computationally efficient architectures, as explained as follows:

- Multimodal Feedback Modalities: Enhancing Perceptual Redundancy:** A clear and consistent trend in recent designs is the move toward integrated multimodal feedback, particularly the synergistic use of audio and haptic cues. These modalities provide perceptual redundancy, which is essential for users navigating dynamic, noisy, or visually ambiguous environments. For example, authors in (Honda et al. 2023) introduced a system combining spatialized audio with a belt-mounted haptic array, resulting in a 24% reduction in mental workload, as measured by the NASA-TLX index—a validated scale in cognitive ergonomics research. This evidence strongly supports the thesis that multimodal interfaces reduce cognitive load by offloading information across different sensory channels, a concept reinforced by recent human factors and neuroergonomics literature findings.
- Ergonomic Design: Balancing Portability and Wearability:** Ergonomics emerges as a key determinant of long-term usability and acceptance. The reviewed systems converge on a weight threshold between 420 and 580 g, aligning with empirical studies that suggest this range as the upper limit for daily wearable comfort. For instance, Hu et al. (Hu et al. 2024) emphasized a lightweight design achieved through hardware offloading, which **achieves 15.8% better than baselines and improves pose tracking accuracy by 25% through real-time audio feedback** (Hu et al. 2024). In contrast,

Patel et al. (Patel et al. 2020b) adopted a chest-mounted, hands-free configuration, which offered less distraction but introduced a higher learning curve, possibly due to the altered interaction paradigm. This trade-off underscores the necessity of ergonomic co-design, ensuring that innovations in form factor do not compromise intuitiveness or accessibility.

- Smartphone Integration: Offloading Computation, Enhancing Interaction:** One of the most transformative trends in modern smart cane systems is the integration of smartphones as both computational co-processors and user interface hubs. This evolution marks a shift from monolithic, hardware-constrained devices to distributed, intelligent ecosystems that leverage mobile platforms for advanced perception, planning, and feedback delivery. Modern smartphones are equipped with Neural Processing Units (NPUs) that allow real-time AI inference with low latency and energy consumption. As shown in Fig. 7, these capabilities are reflected in the top layer—Mobile Application—which handles: Personalization (user preferences, training modules), Route Planning (dynamic updates, real-time rerouting), and Connectivity (integration with cloud services, emergency contact, and caregiver monitoring). Recent works, such as Hu et al. (2024) demonstrate the use of ARCore-enabled offloading for simultaneous localization and object recognition, enabling SLAM execution on-device while minimizing the cane's onboard computational load (Hu et al. 2024). Meanwhile, other implementations, such as Díaz-Parra et al. (2023) and Honda et al. (2023), integrate smartphones as adaptive user interface controllers, utilizing voice commands, environmental annotations, and accessibility APIs to improve user engagement and reduce training overhead (Díaz-Parra et al. 2023; Honda et al. 2023). These solutions capitalize on the ubiquity and familiarity of smartphones, reducing the need for specialized training

Fig. 7 Edge-AI and Federated Learning architecture in a smart cane system. Real-time inference and localized learning are performed at the device level, while preserving privacy and reducing latency



and increasing adoption likelihood. A conceptual Fig. 6 showing how human interaction layers are embedded in the smart cane architecture.

Figure 5 provides a novel visual framework that captures this systemic layering of HCI and processing components in smart cane architectures. The figure is not merely illustrative but reflects a functional modular design, where each layer contributes to the system's intelligence, adaptability, and usability:

- i. **Ergonomic & Sensor Design:** Forms the foundational hardware layer, ensuring that devices are wearable, portable, and capable of accurate environmental perception via LiDAR, IMU, or depth sensors.
- ii. **Multimodal Feedback:** Enables real-time interaction through audio and haptic channels. The framework introduces context-appropriate feedback, dynamically choosing the feedback modality and intensity based on environmental and user-specific variables (e.g., ambient noise, user stress level, or lighting).
- iii. **AI & SLAM Processing:** Serves as the intelligent core of the system, performing obstacle detection, semantic scene interpretation, and optimal navigation path computation. Smartphone offloading reduces on-board processing complexity while retaining real-time responsiveness.
- iv. **Mobile Application Layer:** Enables high-level interaction, such as remote training, user configuration, adaptive learning, and connectivity with external systems (e.g., emergency alerts, caregiver dashboards).

The novelty of Fig. 7 lies in its structured integration of HCI principles with sensor-driven navigation, forming a holistic model for next-generation smart canes. Rather than treating perception, processing, and interaction as isolated units, this

framework synthesizes their interaction to support a low-cognitive-load, high-adaptability navigation aid. Furthermore, by explicitly highlighting smartphone integration, the framework provides a scalable and economically viable direction for future development, minimizing proprietary hardware while maximizing user familiarity and customization. In summary, Fig. 7 transforms the concept of smart canes from single-purpose tools to user-centered assistive ecosystems, enabling safer, smarter, and more adaptive mobility through layered and interconnected AI-HCI integration.

5.4 Discussion

The success of smart canes is not exclusively determined by technological advancement; success is dependent upon the integration of human–computer interaction principles, including cognitive load management, ergonomic design, and adaptive user experience. It has been shown that multimodal feedback (e.g., auditory, haptic, visual) increases the user's situational awareness and distributes the burden of perception, but it must be structured hierarchically in order to avoid sensory overload. Recent empirical findings, such as those by Skovfoged et al. (2022a, b), demonstrate that feedback modality significantly affects task completion time and user stress levels. Their study found that combining tactile and auditory cues improved navigation speed and accuracy while minimizing cognitive strain, especially in dynamic or noisy environments. This underscores the importance of designing context-aware and user-adaptive feedback strategies (Skovfoged et al. 2022b). To address practical deployment challenges and user-centric design considerations, Table 4 provides a synthesized summary of real-world evaluations and HCI-focused findings from recent studies (2021–2024), including cognitive load metrics, navigation accuracy, and usability feedback. There is an urgent need for adaptive feedback models that are responsive to context

and user states. Additionally, ergonomics plays a key role: lightweight, intuitive, and well-balanced designs facilitate user adoption and reduce fatigue. In spite of the promise of hands-free and wearable solutions, these products must align with the users' natural movements and sensory expectations. A new generation of architectures allows AI-intensive tasks (e.g., SLAM, object recognition) to be offloaded to smartphones, leveraging on-device NPUs for real-time processing. This shift transforms smartphones from accessories into cognitive companions, enabling personalization, connectivity, and energy-efficient operation. Ultimately, smart cane systems must be co-designed with visually impaired users to ensure alignment with real-world needs and contexts. As technology becomes more sophisticated, designers must prioritize usability, comfort, and cognitive compatibility, ensuring that inclusive innovation remains at the forefront.

6 Comparative analysis of smart cane technologies

As smart canes evolve from simple tactile aids into intelligent navigation systems, it becomes essential to evaluate their performance not only in isolation but also comparatively across standardized criteria. This section addresses the need by offering a unified, metric-based synthesis of current smart cane technologies, including sensing modalities, processing strategies, and user interaction methods. Table 5 provides a side-by-side comparison of five representative methods commonly used in smart canes: ultrasonic sensors,

LiDAR, YOLOv5 with visual SLAM, smartphone-based AI co-processing, and wearable haptic belts. Each method is assessed across ten standardized metrics that reflect engineering performance, usability, and real-world feasibility:

- **Cost:** Ranges from Low to High depending on component availability and integration complexity.
- **Accuracy:** Describes precision in obstacle detection and localization.
- **Power Efficiency:** Indicates suitability for battery-powered wearable systems.
- **Real-Time Capability:** Refers to responsiveness during active navigation.
- **Indoor vs. Outdoor Use:** Highlights environmental suitability.
- **User Acceptance:** Captures ease of use, comfort, and social acceptability.
- **Cognitive Load:** Evaluates the mental effort required to interpret system feedback.
- **Portability:** Considers weight, size, and wearability factors.
- This matrix-based analysis serves multiple purposes:
- It synthesizes scattered discussions found earlier in the paper into a cohesive format.

It highlights trade-offs between systems. For instance, LiDAR provides high accuracy but suffers in terms of power consumption and portability, while smartphones with AI co-processing balance accuracy and usability at a lower cost. It

Table 5 Comparative matrix of smart cane technologies based on standardized evaluation metric

System/ Study	Sensor Fusion	AI/SLAM Capa- bility	Feedback Mechanism	Indoor Adaptability	Outdoor Adaptability	Energy Efficiency	Cost Scalability	Overall Integration
WeWALK (Suman et al. 2022)	Moderate (Ultra- sonic + Phone)	Limited (Navi- gation App only)	Voice + Vibra- tion	Low (No indoor SLAM)	High (GPS- based)	High (Battery- based)	Moderate	Partial
SLAM- Cane (Bamdad et al. 2024)	High (RGB-D, IMU, Ultra- sonic)	Strong (Custom SLAM)	Voice Feedback	High	Medium	Medium	Low (Costly sensors)	High
Infra-Cane (Mai et al. 2023)	Low (Infrared only)	None	Basic Beep	Low	Medium	High	High	Low
AI-Cane (Zafar et al. 2022)	High (Sen- sor + IoT)	Moderate (Pre- trained models)	Adaptive Audio	Medium	High	Medium	Moderate	Medium
Proposed Frame- work	High (Multi- modal fusion)	High (SLAM + ML)	Multilingual Voice + Hap- tics	High (BLE & SLAM)	High (GPS/5G)	High (Solar + Edge)	High (Modular Design)	Very High

emphasizes the need for design trade-offs, such as selecting medium-accuracy wearable haptic belts that offer high portability and user acceptance, versus deploying advanced YOLO-based systems that demand more computational resources but deliver high recognition fidelity. Through this comparative framework, developers and researchers can quickly identify design sweet spots and limitations, helping guide future engineering and user-centric development. A proposed framework is explained in Section 7. The matrix also sets the foundation for future quantitative benchmarking, where specific real-world metrics such as latency (*ms*), frame rates (FPS), or energy consumption (*mAh*/hour) can further enrich the evaluation.

6.1 White cane: baseline assistive tool (Pre-2000s)

The white cane is considered to be the most basic mobility assistance technology, relying only on tactile feedback and manual navigation (Helmy et al. 2011a). In spite of its low cost and simplicity, the device does not provide automated perception or environmental awareness, and it does not provide any support from artificial intelligence or digital feedback. Because its functionality is dependent entirely on the user's ability to detect nearby surfaces, elevated obstacles, low-hanging barriers, or moving hazards may not be detected (Helmy et al. 2011b).

6.2 Ultrasonic cane: introduction of basic sensing (2000s–2015)

Ultrasonic canes introduced short-range obstacle detection, marking the transition from mechanical to electronic assistive systems (Helmy et al. 2011a). Using basic GPS for location approximation and audio/vibration feedback, these devices offered incremental improvements in environmental awareness, especially for detecting close-range obstructions. However, their narrow detection range, susceptibility to signal noise, and lack of semantic understanding limit their performance in unstructured or dynamic environments (Daudpota et al. 2017). These canes cannot distinguish between types of obstacles (e.g., wall vs. moving object), and their feedback is often generic, requiring users to interpret vague cues. Additionally, GPS-only navigation suffers from reduced accuracy in indoor or urban canyon settings, making ultrasonic canes insufficient for comprehensive navigation (Sibu et al. 2023).

6.3 LiDAR-based cane: precision mapping and obstacle avoidance (2016–2021)

LiDAR integration brought a significant advancement in obstacle detection by enabling high-resolution 3D

environmental mapping and SLAM capabilities (Zhang et al. 2016). These systems support real-time localization, dynamic obstacle avoidance, and accurate path planning. The major improvement lies in precision and spatial awareness, allowing users to navigate complex terrains, avoid collisions, and maintain awareness of their relative position in space (Asati et al. 2019). However, these benefits come at a cost: LiDAR systems are typically expensive, power-hungry, and require bulky sensors, raising issues of portability and accessibility, particularly for low-income or elderly users (Romlay et al. 2023). Moreover, the feedback mechanisms—primarily voice and haptic cues—must be carefully calibrated to prevent information overload, a known issue in high-density sensor environments.

6.4 AI Smart cane: context-aware and adaptive systems (2021–Present)

The latest generation of smart canes incorporates computer vision, deep learning, and sensor fusion, combining SLAM with GPS to offer both semantic and spatial navigation (Kumar et al. 2024). These canes leverage YOLO-based object detection, CNNs, and even reinforcement learning for predictive path planning (Sibu et al. 2023). Feedback mechanisms have also evolved to multimodal formats, such as app-based interfaces, haptic feedback, and voice instructions, enabling adaptive user engagement based on cognitive load, user preference, or environmental conditions. Importantly, smartphone integration allows for on-device AI processing or cloud offloading, making these systems scalable and customizable. Despite their advantages, AI smart canes face challenges in energy efficiency, real-time processing demands, and system reliability in edge cases (e.g., fog, poor lighting, occlusion). The requirement for continuous updates, calibration, and AI model retraining can be burdensome and may introduce usability barriers for non-tech-savvy users.

6.5 Discussion

The evolution of assistive canes reflects a shift from simple, user-dependent tools to intelligent navigation systems. Traditional white canes, while affordable and reliable, offer limited assistance in complex environments. Although traditional white canes are reliable and reasonably priced, they provide limited assistance when faced with complex environments. Although ultrasonic canes introduced basic sensing, they did not provide semantic interpretation. LiDAR-equipped canes allowed for spatial mapping and SLAM-based path planning, but presented challenges in terms of cost, power consumption, and feedback complexity. AI-driven smart canes incorporate deep learning, computer vision, and multimodal sensors to provide contextual

awareness and adaptive navigation. Nevertheless, their sophistication results in a number of trade-offs, including computational requirements, energy consumption, and user accessibility concerns. The success of smart canes will depend on balancing advanced functionality with usability to ensure that they empower users without causing cognitive burdens.

7 Challenges and future research directions

Even though smart cane technologies and assistive navigation systems have made promising advances, several critical challenges remain that hinder their scalability, accessibility, and real-world acceptance. The challenges are not limited to the technical domain but also encompass a number of economic, social, ergonomic, and regulatory concerns, reflecting the complexity of designing inclusive and sustainable assistive solutions for individuals who are visually impaired. Smart canes have certainly been enhanced by the integration of AI, SLAM, multi-modal sensor fusion, and IoT-based connectivity; however, these enhancements often increase system complexity, financial burden, and usability issues. As part of this section, we analyze the key limitations currently faced by the field, including the high costs of advanced sensor-integrated smart canes, their limited deployment in real-world scenarios, and their lack of standardized protocols for navigation and interoperability. As a result of these issues, smart cane systems cannot be widely adopted and implemented effectively in diverse environmental and sociocultural contexts. Furthermore, recognizing the rapidly evolving landscape of assistive technology, we present a forward-looking perspective on future research directions that aim to bridge the identified gaps. As outlined in Sect. 8.2, we propose strategic pathways encompassing technological innovation, functional scalability, and socio-ethical alignment. These directions advocate for the adoption of next-generation wireless networks (5G/6G), edge-cloud intelligence architectures, adaptive and modular hardware design, and international standardization efforts. Collectively, they seek to foster the development of robust, affordable, and user-centric smart mobility solutions that can be effectively deployed across global contexts. The following subsections provide a detailed exploration of the current challenges (Section 7.1) and articulate transformative research agendas for the future (Sect. 8.2), thereby contributing to the long-term vision of accessible and intelligent assistive navigation systems for the visually impaired.

7.1 Challenges

- i. **High Cost of Advanced Smart Canes:** A number of cutting-edge technologies have been integrated into

smart canes, including LiDAR, GPS, ultrasonic sensors, camera modules, and haptic feedback systems. These innovations have significantly enhanced navigational capabilities for visually impaired individuals. However, the integration of such advanced components dramatically escalates the bill of materials, which directly impacts the affordability and scalability of these systems. For instance, LiDAR modules alone can account for up to 40% of a smart cane's production cost (Jain and Patel 2023). The primary user demographic often faces **significant socioeconomic challenges**, making these devices financially inaccessible without external subsidies or public funding (Kumar et al. 2024). **Moreover, the total cost of ownership is further increased by the need for frequent calibration, firmware updates, and technical maintenance, especially in sensors that degrade due to environmental exposure.** This issue is amplified in low-resource settings where technical support is scarce or non-existent. In terms of policy and engineering, it is imperative to explore **cost-efficient, modular, and sustainable design paradigms** that preserve core functionality without sacrificing usability. This includes leveraging open-source, community-supported hardware platforms (e.g., Arduino, Raspberry Pi) and employing minimalist principles such as feature pruning and adaptive sensing, where the system dynamically activates sensors based on environmental context (Remya et al. 2023). Additionally, public-private partnerships could catalyze subsidized production pipelines through interdisciplinary collaborations, particularly between government agencies, non-profits, and private manufacturers. These collaborations should also explore circular economy models, where used smart canes can be refurbished or upgraded at minimal cost to reduce electronic waste and support environmental sustainability.

- ii. **Limited Real-World Deployment and User Adoption:** Despite encouraging results in laboratory and controlled environments, smart canes are not widely distributed in real-world scenarios. This gap has been attributed to several factors, including a lack of user-centered design methodologies, insufficient longitudinal trials, and the absence of contextual adaptation to diverse urban terrains and cultural norms (Kumar et al. 2024). **Many prototypes fail to transition into practical, daily-use tools because they are designed without iterative feedback from end users. Aesthetic bulkiness, high cognitive load from simultaneous feedback modalities, short battery life, and non-intuitive interfaces are recurrent design flaws that degrade usability and acceptance.** Studies have shown that smart devices requiring more than

two hours of training or daily charging are 35–50% more likely to be discontinued by users within the first month (Tian et al. 2024). Beyond technical barriers, psychosocial stigma plays a significant role in deterring adoption. A conspicuous assistive device may unintentionally signal disability, reinforcing negative stereotypes or attracting unwanted attention in public. This stigma is often heightened in conservative or visually normative cultures, where disability is underrepresented or stigmatized in media and public discourse. Future systems must incorporate inclusive and participatory design approaches to overcome these barriers. A variety of co-design workshops, ethnographic studies, and multi-stakeholder evaluations are involved, including visual impairment users, clinicians, caregivers, and planners. Developers will be able to create devices that are technically sound, socially acceptable, culturally relevant, and emotionally empowering only if they engage in a sustained collaboration throughout the design lifecycle.

- iii. **Lack of Standardized Navigation Protocols:** Smart canes continue to face a serious bottleneck in the integration and interoperability of communication and navigation protocols. It is important to note that current solutions rely heavily on proprietary firmware, bespoke navigation stacks, and vendor-specific communication layers, which lead to fragmented ecosystems and compatibility problems. There is a lack of standardization in this field, which hinders the seamless integration of smart urban infrastructure, including tactile paving, IoT-enabled pedestrian signals, and alerts regarding public transportation in real time. Also, due to inconsistent GPS accuracy, uncalibrated local maps, or unlicensed radio frequencies, performance varies across regions, compromising both the safety and reliability of the guidance. Furthermore, assistive navigation systems have very little oversight from government agencies, with no enforcement of precision requirements, safety margins, or data privacy laws. Regulatory gaps such as these raise serious concerns, particularly as smart canes become edge-computing platforms capable of collecting, processing, and potentially transmitting data regarding location and behavior. Consequently, the community must advocate for standards that have been ratified internationally, and which have been guided by the International Standards Organization (ISO), the Institute of Electrical and Electronics Engineers (IEEE), and the Web Content Accessibility Guidelines (WCAG). There should be benchmark datasets, unified API interfaces, frameworks for evaluating sensor accuracy, and checklists for complying with privacy and cybersecurity laws. Similarly, the development of

an open, federated testing environment, which allows for the evaluation of smart canes in controlled and diverse real-world simulations, could accelerate the progress towards certification and mass adoption.

- iv. **Technology Transfer and Commercialization Barriers:** One of the critical challenges in the advancement of smart cane systems is the limited translation of academic prototypes into deployable commercial solutions. Research systems often emphasize advanced AI (e.g., SLAM, AR guidance), yet lack the robustness, cost-efficiency, and ergonomic polish required for large-scale deployment. Conversely, commercial products prioritize user simplicity and cost, but frequently omit AI-driven features due to power or budget constraints.

Table 6 presents this contrast in a structured format, comparing sensor types, AI integration, localization capabilities, feedback mechanisms, connectivity, power sources, usability, cost, and availability. This comparative overview highlights the technology transfer gap between academic innovation and real-world deployment, identifying opportunities for future research to focus on miniaturization, energy efficiency, cost optimization, and user-centric design. Bridging this gap is crucial to enable the next generation of smart canes that offer both intelligence and practicality for visually impaired users.

7.2 Future research directions

The development of mobile computing, communication technologies, and AI is accelerating the development of assistive navigation systems at an unprecedented speed. There are several critical challenges that remain in the development of robust, secure, scalable, and context-aware smart cane ecosystems, despite substantial progress. The objectives of this section are to bridge existing gaps and to develop smart mobility solutions for visually impaired individuals that are future-proof and flexible. As shown in Fig. 6, the enhanced architecture of a smart cane system incorporates future research directions. The system consists of a microcontroller that collects and processes data from multiple sensors, including audio, LiDAR, IMUs, GPS, and a camera. An adaptive algorithm-based ML module supports intelligent decision making (Alslaity and Orji 2024). A multilingual voice-interfaced mobile app facilitates seamless user interaction with indoor localization through Bluetooth Low Energy (BLE). To support health monitoring, usage analytics, and system diagnostics, data is securely encrypted and transmitted to a cloud server. The system leverages solar energy for sustainable operation and integrates next-generation connectivity (5G/6G) to enable real-time communication. Within the broader context of a

Table 6 Comparison of commercial vs. research smart cane systems

Criteria	WeWALK (Commercial)	UltraCane (Commercial)	Mai et al. (2024, Research Prototype)	Panazan and Dulf (2024, Research Prototype)
Sensor Types	Ultrasonic, IMU	Dual Ultrasonic	2D LiDAR, RGB-D camera, IMU, GPS	Depth camera, Ultrasonic, IMU
Obstacle Detection Range	~2–3 m	~3 m	~6–8 m (multi-sensor fusion)	~4–6 m
AI Integration	Basic voice assistant (limited AI)	None	YOLOv5 for object detection, semantic mapping	CNN-based semantic SLAM
Indoor Localization	Not supported	Not supported	Visual SLAM + IMU	Visual-inertial mapping
Multimodal Feedback	Voice + vibration feedback	Tactile feedback (vibration motors)	Voice, dynamic haptic, mobile app interface	Smartwatch + voice + vibration
Connectivity	Bluetooth, Mobile App	No connectivity	Cloud + GPS + Mobile Network	IoT via Wi-Fi/BLE
Power Source	USB rechargeable	Replaceable batteries	Rechargeable battery + solar support	Rechargeable battery
Ease of Use	High (intuitive app and TTS interface)	Medium (training needed)	Medium (calibration + AI model training required)	Moderate (for tech-savvy users)
Estimated Cost (USD)	~\$500	~\$1,000	Not commercialized, est. > \$800	Not commercialized, est. > \$700
Market Availability	Available globally	Available in selected markets	Laboratory prototype – not commercially available	Research use only

smart city (Rodrigo-Salazar et al. 2021), this architecture supports intelligent urban mobility by interfacing with connected infrastructure, such as adaptive traffic signals, smart crosswalks, and location-aware public services, thereby enhancing situational awareness and safety for visually impaired users.

i. **Real-Time Intelligence via 5G/6G and Edge-Cloud**

Collaboration: The rollout of 5G and early-stage 6G infrastructure offers low-latency, high-throughput networks capable of supporting real-time smart cane applications. Offloading computationally intensive tasks—such as semantic SLAM, environmental segmentation, and crowd-aware path planning—to edge-cloud servers can significantly reduce on-device resource constraints (Naved et al. 2023). More importantly, cloud-connected systems enable dynamic content updates such as construction detours, ambient crowd density, and dynamic crosswalk signal statuses. A key direction is the development of resilient cloud-edge orchestration protocols. These should support context-aware fallback mechanisms—e.g., switching to edge AI inference when bandwidth degrades—and integrate continuous environmental learning, enabling real-time co-navigation and hazard broadcasting. Furthermore, seamless integration with Mobility-as-a-Service (MaaS) ecosystems can allow visually impaired users to coordinate public transport, elevators, and navigation via a unified interface (Hauslbauer et al. 2024; Li et al. 2024). Future research should

focus on robust cloud-edge orchestration algorithms, ensuring seamless failover to local AI when connectivity drops.

ii. **Edge AI and Federated Learning for On-Device**

Adaptability: Even though cloud computing can enhance computational capabilities, it may not be appropriate in environments with intermittent connectivity or a critical latency constraint. The field of edge AI, which involves performing AI inference directly on the device, has emerged as a promising approach to address this issue (Veerasamy et al. 2023). As a result of edge artificial intelligence, smart canes are able to make instantaneous decisions based on local sensor data, such as detecting nearby obstacles, identifying traffic signals, or predicting the path of pedestrians and vehicles (Veerasamy et al. 2023). Due to recent advances in lightweight AI models and energy-efficient hardware (e.g., Google Coral TPU, NVIDIA Jetson Nano), it is now possible to deploy real-time neural networks on portable devices (Jivrajani et al. 2023). Federated Learning (FL) presents an especially promising research frontier. By enabling collaborative model training across decentralized smart canes, FL ensures that user data never leaves the device, preserving privacy while improving generalization. Future work should examine federated distillation, adaptive aggregation, and communication-efficient FL protocols, especially under constrained energy and bandwidth conditions (Kapoor and Kumar 2024). Figure 7 illustrates an FL-enabled edge AI architecture sup-

porting this vision. Thus, navigation ecosystems can be personalized and secure by learning from broader usage patterns without exposing individual information. Furthermore, as mentioned in (De Mel et al. 2024), it can offload some tasks from the cloud to the edge to maintain the balance of the network, which can avoid communication delay and insecure transmission, and improve the device's overall performance. In addition, it can expand more functions based on the IoT, such as real-time monitoring of the location and the surrounding environment of the VIP (Naved et al. 2023).

- iii. **Personalized Human-Centered AI:** Navigation needs are highly individual. Age, cognitive style, risk tolerance, and sensory preference vary across the visually impaired population. Human-centered AI (HCAI) enables adaptive systems that tailor feedback and path planning to user profiles. For example, biosignal integration (heart rate, galvanic skin response) can trigger emotion-aware assistance, offering more verbose instructions during stress or switching to haptic mode in noisy areas (Kuriakose et al. 2023): Key research directions include:

- Multimodal adaptation models: context-driven switching between audio, haptic, and visual feedback.
- Behavioral cloning & imitation learning: enabling systems to mimic users' preferred paths over time.
- Assistive co-pilots: voice-interactive agents that can clarify instructions, anticipate confusion, and assist with decision-making.

- iv. **3D Spatial Understanding with Advanced Sensor Fusion:** While current systems detect ground-level hazards effectively, they struggle with vertical and transient obstacles like overhangs, escalators, or rotating doors (De Mel et al. 2024). For example, many devices struggle to perceive head-level obstructions (e.g., signs, overhanging branches) and 3D spatial anomalies such as spiral staircases or escalators, which require a holistic understanding of height, depth, and structure. The next frontier involves combining high-resolution LiDAR, depth cameras, and inertial sensors into a unified perception stack. Future systems should leverage: This includes the integration of high-resolution 3D LiDAR (16 or 32-line TOF sensors) with RGB-D cameras and IMUs for pose correction. The combination of dense point cloud data, color-depth cues, and motion feedback can offer a more accurate reconstruction of complex scenes. Key technical advancements include: 1) Dynamic multi-frame inference to reduce occlusion. 2) Joint sensor

calibration pipelines for time-synchronized fusion. 3) NeRF for fine-grained, view-invariant 3D reconstruction—especially for indoor scenes with rich geometry (Theodorou et al. 2023). 4) Vertical semantic SLAM, capable of representing 3D layouts across multilevel buildings. Such developments would unlock advanced applications such as context-aware elevator detection, multi-story wayfinding, and hazard anticipation in crowded environments.

- v. **Generalizable SLAM and Object Detection in Real-World Conditions:** SLAM systems today struggle with occlusions, dynamic scenes, and unusual terrain (e.g., snow, gravel, or puddles). Similarly, vision-based object detectors underperform in low lighting or with deformable obstacles. Research should focus on: 1) Self-supervised SLAM pipelines that adapt to new environments with minimal annotations. 2) Angle-invariant and texture-agnostic detection, enabled by spatial transformers and attention networks. 3) Few-shot learning and continual learning, to personalize and update models over time. 4) Generative replay for knowledge retention across diverse geographies or climates. Such techniques would allow smart canes to generalize effectively, support lifelong learning, and accommodate domain adaptation without catastrophic forgetting.
- vi. **Sustainability and Open Innovation:** Smart canes must move beyond performance and embrace sustainability, affordability, and user maintainability for widespread adoption. Prolonged battery life is essential for daily use, especially in resource-limited settings. Energy efficiency, cost, and repairability are crucial for large-scale deployment. Future research should investigate: 1) Dynamic energy scaling via adaptive inference rates and duty-cycling of sensors. 2) Eco-friendly materials and modular casing designs to facilitate upgrades and recycling. 3) Open-source SDKs and APIs that allow researchers, startups, and NGOs to extend functionality and localize tools. These efforts would democratize innovation and allow communities to co-develop localized smart cane solutions aligned with regional needs.
- vii. **Standardization, Governance, and Policy Frameworks for Smart Cane Navigation:** Technological development alone is insufficient—policy and infrastructure must evolve in parallel to ensure equity, access, and interoperability. Critical areas of focus include: 1. Implementing global standards for HCI features, such as the consistency of voice commands, the level of vibration feedback, and the timing of auditory notifications. 2. Establishing communication protocols to ensure seamless operation across Bluetooth Low Energy (BLE), 5G/6G, and edge cloud infrastruc-

ture; this is essential for real-time updates and remote monitoring. 3. Providing subsidies or insurance coverage to vulnerable populations to reduce cost barriers associated with accessibility compliance. Developing cane-compatible pathways requires collaboration with urban planners, partnering with disability advocacy groups to conduct user testing, and collaborating with governments to embed smart cane systems into national transportation and health programs.

- **Fragmentation and Integration Challenges:** Smart cane systems continue to face significant challenges due to the absence of standardized communication and navigation protocols. As current devices often operate on proprietary firmware, custom navigation stacks, and vendor-specific communication protocols, this has led to a fragmented ecosystem. Such fragmentation impedes seamless integration with broader smart urban infrastructure, including tactile paving grids, IoT-enabled pedestrian signals, and real-time public transport alerts. Moreover, variability in GPS accuracy, uncalibrated indoor maps, and the use of unlicensed radio frequencies further undermine the consistency and reliability of these systems across regions. This inconsistency not only affects system performance but also compromises user safety and trust.
- **Overview of Existing Standards:** A critical discussion of existing standards reveals a partial but promising foundation upon which to build more robust, integrated systems. ISO 9241–210, which emphasizes human-centered design, is particularly relevant for ensuring usability and accessibility in smart cane interfaces. For location tracking, ISO/IEC 24730 provides useful guidelines on real-time location systems (RTLS) that can be extended to assistive navigation. Similarly, IEEE P2040, addressing wearable consumer electronics, can serve as a basis for establishing compliance and integration parameters for smart canes. IEEE 802.15.4 is especially relevant for low-power wireless communication, such as Bluetooth Low Energy and Zigbee, which are commonly used in these devices. Though originally designed for digital content, WCAG 2.1's principles of perceivability and operability provide valuable guidance for designing multimodal feedback systems.
- **Gaps in Regulatory and Ethical Governance:** Despite these existing frameworks, there is no unified standard that fully addresses the unique requirements of smart canes, especially those operating as edge-computing platforms. Regulatory oversight in this area remains minimal, with few enforceable guidelines related to precision,

safety margins, or user data privacy. This is particularly concerning given the increasing ability of smart canes to collect, process, and potentially transmit sensitive location and behavioral data. Without proper governance, these capabilities raise ethical and legal questions that must be urgently addressed.

- **Path Toward Unified International Standards:** To move forward, it is essential to advocate for comprehensive international standards. Rather than relying on isolated compliance efforts, there should be a concerted push towards developing open-source, modular APIs that support sensor integration, feedback management, and localization across various hardware platforms. Such standardization would enable greater interoperability between smart cane systems and the urban infrastructure they interact with. Simultaneously, the creation of benchmark datasets and standardized evaluation metrics would facilitate objective assessment of navigation accuracy, latency, and user experience across different environments and use cases.
 - **Role of Policy and Federated Testing:** Moreover, collaboration with government agencies, industry stakeholders, and international standards organizations is necessary to establish certification frameworks for these devices. These certifications should ensure not only functional performance but also compliance with data protection regulations such as the General Data Protection Regulation (GDPR). In this regard, federated testing environments—enabling the assessment of smart canes in diverse simulated and real-world settings—can serve as a critical step toward mass adoption. These controlled trials can offer vital insights into system behavior under various urban conditions, supporting iterative refinement and certification processes.
 - **Inclusive and Scalable Future Outlook:** As smart canes evolve into multifunctional platforms with integrated edge AI, cloud connectivity, and context-aware services, the need for robust policy and standardization frameworks becomes even more urgent. These frameworks must be developed collaboratively and guided by inclusive principles, ensuring they reflect the diverse needs of users with visual impairments. A systematic, globally ratified approach to standardization will not only accelerate innovation but also ensure safe, scalable, and equitable access to smart navigation technologies.
- viii. **Security, Authentication, and Privacy Preservation in Smart Cane Ecosystems:** In parallel with the advancement of smart cane technology into cloud-

connected, AI-driven, and BLE-enabled ecosystems, there has been an increase in cyber-physical vulnerabilities (Tsantikidou et al. 2025). These devices now routinely process sensitive and mission-critical data using real-time edge analytics and contextual localization. As a consequence, the protection of security, privacy, and data integrity has become a central concern in order to ensure that these assistance technologies are trustworthy, usable, and long-term viable (Ashrif et al. 2024a, b, c). Securing next-generation smart cane systems involves a number of critical challenges and potential research.

- Secure Device Authentication and Adaptive Access Control:** Smart canes often serve as both navigational tools and context-aware perceptual systems, continuously logging user movement, environmental context, and interaction patterns. The unauthorized access or manipulation of this data could lead to 1) User tracking and behavioral profiling jeopardize physical safety and autonomy, 2) Environmental spoofing, where malicious actors inject false obstacles or hazards, 3) Remote tampering or firmware compromise, potentially disabling safety-critical functions. A smart cane system is typically resource-limited, with strict restrictions on computational power, memory, and energy. As a result, traditional public-key authentication protocols are often unsuitable. Rather, recent work, such as that by (Ashrif, et al. 2024a, b, c; Ashrif et al. 2025) demonstrates the viability of lightweight mutual authentication protocols using nonce freshness, HMAC-based verification, and optimized key agreement to defend against replay, impersonation, and MITM attacks. For BLE communications between the smart cane and edge devices, these approaches minimize latency and preserve battery life. In particular, this protocol is ideal for Bluetooth Low Energy (BLE) communication between smart canes and companion smartphones or edge devices, where frequent connections may occur as the user moves between different network environments. Key directions include Elliptic Curve Cryptography (ECC) and lightweight challenge-response protocols, tailored for ultra-low-power embedded platforms (Khan et al. 2018). Blockchain-enabled decentralized identity (DID) architectures allow verifiable, tamper-proof identities across heterogeneous devices without centralized trust anchors (Alshamari and Singh 2025). Biometric modalities—such as handgrip pressure signatures or on-device voice-print recognition—to restrict usage to the verified user, with minimal cognitive burden (Hathaliya et al. 2020). Additionally, Policy-Based Access Control

(PBAC) systems can dynamically adjust user privileges based on time, location, and behavioral history, ensuring only legitimate operations are permitted under defined circumstances (Ayedh et al. 2023).

- Resilient Communication Protocols for Edge-Cloud Architectures:** Given their reliance on BLE, Wi-Fi, and increasingly, 5G/6G, smart canes are vulnerable to classic network-layer attacks, including eavesdropping, spoofing, replay, and man-in-the-middle (MITM) intrusions. These vulnerabilities are amplified in public or semi-trusted environments such as transit systems, shopping malls, or urban plazas. Future work should focus on developing contextual encryption pipelines that dynamically adapt to environmental risk profiles and network trust levels. Critical advances include post-quantum cryptography algorithms (e.g., lattice-based key exchange) to secure communication against quantum adversaries. Zero Trust Architectures enforce continuous identity verification and behavioral monitoring for every node-to-node interaction. Context-aware encryption schemes, where factors like data sensitivity, user activity, and proximity to untrusted nodes modulate encryption granularity and frequency. Moreover, adversarial perturbation techniques—originally designed for privacy-preserving computer vision—could be repurposed to obfuscate sensitive visual and audio streams, safeguarding non-user entities (e.g., bystanders) inadvertently captured during real-time operation.
- Privacy-Preserving Learning and Collaborative Intelligence:** Integrating real-world AI inference in smart canes introduces novel privacy risks, particularly around surveillance and unintentional data disclosure. As models are increasingly trained and refined using user-generated data, the ethical and legal implications (e.g., under the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA)) become non-trivial. Emerging solutions must embrace federated intelligence frameworks, where decentralized smart cane nodes cooperatively learn without raw data ever leaving the device. This not only preserves user privacy but also improves scalability and personalization. Key techniques include FL, where model updates—not data—are transmitted to the central server, with potential enhancements through federated distillation and adaptive aggregation. Differential Privacy (DP)-enhanced FL, ensuring that individual contributions remain indistinguishable within the overall update. On-device anonymization filters for vision and audio streams, enabling safe pre-transmission sanitization. Homomorphic encryption and

secure multi-party computation can facilitate joint inference or mapping without any entity accessing the complete dataset, maintaining data confidentiality and model utility for collaborative tasks such as multi-user SLAM. These methods ensure that smart cane systems can evolve intelligently while maintaining user confidentiality. Moreover, hybrid schemes such as Federated Distillation and Secure Multi-Party Computation (SMPC) enable collaborative AI tasks (e.g., multi-user SLAM or shared obstacle mapping) without compromising user identity or safety.

- Cyber-Physical Security and Tamper Resilience:** Because smart canes are deeply embedded within users' physical navigation loops, their compromise could result in harmful outcomes. Ensuring end-to-end system integrity involves securing both firmware and hardware layers. Tamper-resilient designs include secure boot mechanisms, hardware root-of-trust, and runtime attestation protocols that verify firmware authenticity at every startup. Physical security is reinforced through tamper-evident packaging, sensor-level hash chains, and outlier detection engines that identify spoofed or inconsistent data streams. Recent research supports embedding runtime anomaly detection models on edge devices to detect hardware failures or malicious interference in real-time. These systems can monitor behavioral deviations, context shifts, or signal anomalies that indicate a breach, even in the absence of direct user oversight. The integration of secure boot processes and cryptographically verified firmware updates establishes trust anchors critical for smart canes used in public and dynamic environments. Together, these mechanisms fulfill the broader goal of embedding security, privacy, and resilience at every layer—from physical sensors to cloud inference pipelines—ensuring inclusive and trustworthy smart mobility solutions.

8 Conclusion

Smart cane technology has advanced from basic tactile tools to intelligent, multi-sensor systems that provide real-time navigation, obstacle avoidance, and situational awareness for users with visual impairments. Through this review, we examined how developments in SLAM, machine learning, and sensor fusion have transformed assistive mobility into a more adaptive and intelligent experience. However, despite these advancements, challenges persist—particularly in terms of indoor localization, system energy efficiency, interoperability, and user acceptance. Addressing these gaps,

we proposed a comprehensive architectural framework that integrates edge AI processing, Bluetooth-based indoor positioning, 5G/6G-enabled secure connectivity, solar-powered operation, and cloud-based health monitoring. This layered model is designed to ensure low-latency processing, enhanced data privacy, and long-term autonomy, making it scalable across diverse urban environments. What sets this survey apart is its integrative approach. While prior works have focused on specific components (e.g., AI, SLAM, sensors, LiDAR, cameras, NeRF, or ergonomics), this study unifies these perspectives into a cohesive blueprint for future smart cane systems. By examining the intersection of technological innovation and human-centered design, our survey not only maps current capabilities but also defines strategic directions for achieving inclusive, connected, and sustainable mobility. Going forward, we emphasize the importance of developing lightweight, personalized models; refining ergonomic design; and embedding smart cane systems within broader smart city infrastructure. Through interdisciplinary collaboration spanning engineering, design, and policy, smart canes can evolve from standalone aids into essential enablers of equitable, intelligent urban mobility.

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Data Availability All data supporting the findings of this study are included in the article. No additional data are available.

Declarations

Conflict of interest The authors declare no conflicts of interest.

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